Zarif Sharifovich Tadjibaev, PhD

WORLD'S FIRST

IDEAL SEWING TECHNOLOGY ZARIF 2025

BASED ON A ROTATING LOOPER

THE TECHNOLOGICAL DEAD END OF THE 19TH CENTURY

Why do traditional technologies of lockstitch (type 301) and chainstitch (type 401) make complete autonomy of sewing industries impossible

A fundamental analysis of the irremediable limitations of 19th-century sewing technologies

ZARIF 2025 PLATFORM

Ideal sewing technology ZARIF 2025 + AI + Humanoid robots + Autonomous robotic carts.

Ph.D. (Engineering) ZARIF SHARIFOVICH TADJIBAEV

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THE TECHNOLOGICAL DEAD END OF THE 19TH CENTURY

Why the traditional lockstitch technology (type 301) and chain stitch technology (type 401) make full autonomy of sewing production impossible. A fundamental analysis of the irremediable limitations of 19th-century sewing technologies

⚙ ZARIF 2025 PLATFORM

⚙ ZARIF 2025 Ideal Sewing Technology | 🤖 Humanoid Robots
🧠 Artificial Intelligence | 🚚 Autonomous Robotic Trolleys

The foundation of the future automated sewing factory without personnel

Zarif Sharifovich Tadjibaev

Candidate of Technical Sciences (Ph.D.), Inventor of the ZARIF 2025 Ideal Sewing Technology

US Patent No. 6,095,069

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Tashkent, 2026

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ABOUT THIS BOOK

The volume before you is the first book in the series *ZARIF 2025 — The Platform for Autonomous Sewing Production*. It is the world's first comprehensive guide to the revolutionary ZARIF 2025 sewing technology, which resolves once and for all the 170-year-old problem of automating sewing production. It describes in detail how the rotary looper technology, invented by Dr Zarif Tadjibaev, eliminates the fundamental shortcomings of traditional sewing machines (types 301 and 401), making fully autonomous, unmanned garment production possible.

⚠ PROJECT STATUS (2026)

- The ZARIF 2025 prototype has confirmed the viability of the technology.
- The industrial version is under development (concept complete).
- Commercialisation is planned for 2026–2035.

WHO IS THIS BOOK FOR?

- ✅ Engineers and technologists in the garment industry seeking solutions for the automation of production processes.
- ✅ Factory managers and production directors aiming to increase efficiency and reduce dependence on manual labour.
- ✅ Investors in industrial automation, robotics and artificial intelligence.
- ✅ AI and humanoid robot developers (Tesla, Figure AI, Boston Dynamics) interested in integration with sewing equipment.
- ✅ Digital transformation specialists seeking pathways to Industry 5.0.
- ✅ Students and lecturers at technical universities studying advanced technologies in the textile industry.

WHAT WILL YOU LEARN FROM THIS BOOK?

- 🔹 Why traditional sewing machines (types 301 and 401) are incompatible with robots — a detailed analysis of 11 fundamental defects of the lockstitch and 4 irresolvable problems of the chain stitch.

- ◈ How ZARIF 2025 technology eliminates all these shortcomings — the operating principles of the rotary looper, dual-rotation kinematics, Dry Head architecture and digital control.
- ◈ Three types of integration of humanoid robots with ZARIF 2025 machines — from simple loaders to full sewing automation.
- ◈ The economics of the future — how ZARIF 2025 reduces OPEX by 75%, increases OEE to 85%+ and delivers payback within 11.1 months.
- ◈ The commercialisation roadmap — from prototype to fully autonomous factories by 2035.
- ◈ Investment opportunities — how to become part of a market opportunity worth $400–900 billion over 15 years.

HOW TO READ THIS BOOK

This book clearly and consistently distinguishes between two levels of information:

✓ ALREADY PROVEN BY THE ZARIF 2025 PROTOTYPE

These facts are confirmed by 5,000+ hours of testing on a working prototype:

- Viability of the single-direction stitch-formation principle.
- Speed of up to 5,000 stitches per minute on materials up to 8 mm thick.
- Wide Head architecture: material range 0.1–8 mm without adjustment.
- Zero skipped stitches, zero thread breaks, zero needle breakages.
- Premium aesthetics: the reverse side of the new double thread chain stitch (type 401) is visually identical to the lockstitch (type 301).

◈ ENGINEERING CONCEPT

These solutions are described as a realistic next stage:

- Automatic thread-trimming device (1:1 drawings complete; prototype not yet manufactured).
- 360° circular sewing mechanism (1:1 drawings complete; prototype not yet manufactured).
- Industrial version with Dry Head architecture (requires further development).
- Integration with humanoid robots (being tested on prototypes).

PREFACE BY THE AUTHOR

Dear readers,

I wish to begin this book with a personal admission. Thirty-one years ago, in 1994, I asked myself a question which, as it turned out, could transform the entire sewing industry. At the time I was a simple engineer in Tashkent, and the question was this:

"Is it possible to create a double thread chain stitch without an eye-looper?"

At the time I did not know that this question would mark the beginning of a journey that would lead to the creation of the world's first robot-native sewing technology — a technology capable of finally solving the problem that had eluded the entire industry for 170 years.

In 2000, I received US Patent No. 6,095,069 for a method of forming a double thread chain stitch using a rotary looper. That was only the beginning. Over the subsequent 25 years, I transformed it into the world's first ideal sewing technology: ZARIF 2025.

This book is not merely a technical description. It is an invitation to witness how the sewing industry is at last making the transition from the mechanical principles of the 19th century to the digital, autonomous ecosystem of Industry 5.0.

— Zarif Tadjibaev, Tashkent, 2026

INTRODUCTION

The world sewing industry, with a turnover of $2.36 trillion, remains the last major manufacturing sector where 90% of operations are still performed manually. Whilst automotive manufacturing, electronics and logistics have already transitioned to full or partial autonomy, the sewing shop floor has remained trapped in the technological paradigm of 1846. The reason is not a lack of robots, nor any weakness of artificial intelligence. The reason lies in the sewing machine itself, whose architecture has not changed since Elias Howe invented the lockstitch.

Over the past 15 years, the industry spent more than $220 million on attempts to robotise sewing. Adidas closed its "Speedfactory" plants, SoftWear Automation was unable to scale beyond simple T-shirts, and Sewbo remained a laboratory curiosity. Every project foundered against the same barrier: the traditional sewing machine demands constant human intervention — to change the bobbin, adjust the tension, replace a broken needle, re-thread a snapped thread.

ZARIF 2025 is not an improvement on the old paradigm. It is its complete replacement. At the heart of the technology is a rotary looper that makes two full rotations per needle cycle, eliminating the root causes of all key problems: the bobbin, the eye-looper, the "thread triangle" and critically small clearances. The result is a new type of double thread chain stitch that looks like a lockstitch, stretches like a chain stitch and outlasts both.

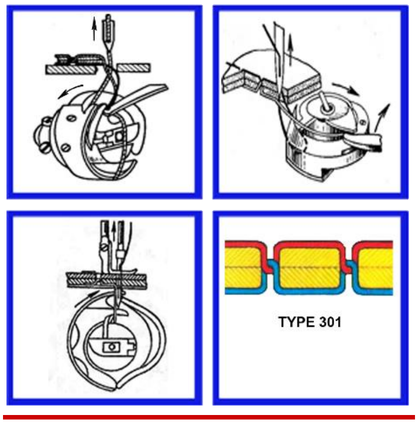
This book will guide you through 31 years of invention history — from the question posed in 1994 to the prototype that, in 2025, proved that ideal sewing technology is possible. And it is already here.

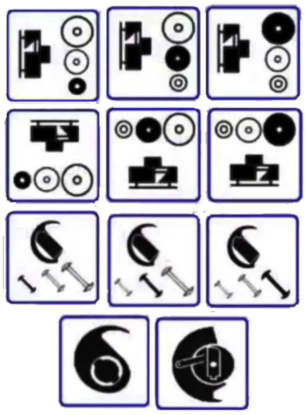
CHAPTER 1. THE CRISIS OF TRADITIONAL TECHNOLOGIES

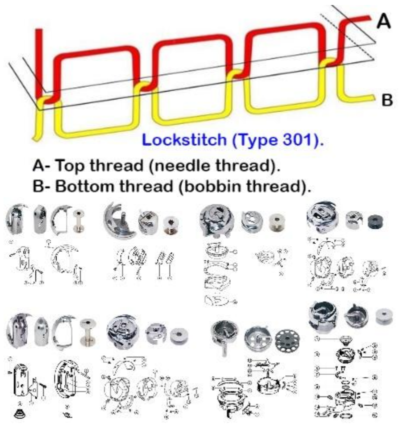
1.1. The Two Pillars That Support the World — and Hold It Back

To appreciate the scale of the ZARIF 2025 revolution, one must first understand the depth of the crisis in which the two dominant technologies of the sewing world find themselves: the lockstitch type 301 (80–85% of the market) and the double thread chain stitch type 401 (15–20% of the market). Both were invented in the mid-19th century and have since been improved only "cosmetically" — servo drives, digital displays, automatic lubrication. But their fundamental mechanical flaws have not been eliminated.

1.2. Lockstitch (Type 301): 11 Irremediable Defects







The lockstitch type 301 (ISO 4915:1991) occupies a dominant position in the world sewing industry — more than 80–85% of all single-needle seams are produced by this method. However, this statistic reflects not technical superiority but the absence of any widely implemented alternative over 177 years of sewing equipment evolution. After 170 years of continuous engineering development, lockstitch technology has reached the physical limits of its operating principle. All modern improvements are engineering compromises that attempt to mitigate the system's inherent shortcomings without solving its fundamental problems.

⚠ KEY CONCLUSION

The eleven fundamental defects described below are NOT the result of engineering errors but are the inevitable consequences of the basic physics of lockstitch formation, embedded in the very DNA of this technology since its invention in the 1840s. They are irremediable within the framework of the existing concept.

DEFECT NO. 1: FREQUENT BOBBIN REPLACEMENT — THE SYSTEMIC BARRIER TO AUTOMATION

The limited quantity of bottom thread on the bobbin requires regular stoppages of the production process. This is one of the principal obstacles to full automation of sewing production, reducing overall efficiency by 10–25%.

Technical Parameters

- A standard industrial bobbin (type L or M) holds 60–110 metres of thread
- At operating speeds of 4,000–5,000 stitches per minute, thread is exhausted within 12–20 minutes
- An average operator performs 30–50 bobbin changes during an 8-hour shift
- Each change: 30–45 seconds of downtime (stop, remove, replace, re-thread, check)
- Total losses: 20–35 minutes per machine per shift

Global Productivity Losses (100 Sewing Machines)

Indicator	Scale of Losses	Equivalent
Daily losses	40–50 hours of productivity	Work of 5–6 operators per shift
Annual losses	~12,000 hours	8–10% of total available working time
When changing thread colour	Downtime increases 2–3 times	Critical in multi-colour production

Attempts to Automate Bobbin Replacement and Their Limitations

There are two types of automatic bobbin replacement systems — however, both solutions are characterised by high cost and significant operational limitations:

System Type	Cost	Critical Shortcomings
Cassette type	$800–2,000	Requires manual cassette replacement every 1–2 hours; frequent failures with thick threads; operator intervention still required
Fully automatic	$2,000–5,000+	Complex construction with its own motor; maintenance costs $200–500/year; frequent breakdowns due to lint; payback period 2–4 years only for large enterprises (>50 machines)

Attempted Solution: Oversized Hooks

To reduce the frequency of bobbin re-threading, manufacturers are compelled to use hooks 2–3 times larger. However, this solution creates a new problem: increased hook size → greater inertia of rotating masses → inability to achieve high sewing speeds (up to 5,000 stitches/min). Large hooks are suitable only for low-speed operations.

> ***Conclusion on Defect No. 1:*** *For 95% of industrial operations, bobbin replacement remains a manual task. The problem is fundamentally irresolvable within the concept of the lockstitch: the bottom thread MUST ALWAYS reside on a bobbin inside the hook.*

DEFECT NO. 2: CRITICAL LOSS OF TOP THREAD STRENGTH

Progressive mechanical degradation of the thread occurs because the same section of top thread passes repeatedly through the narrow apertures of the mechanisms and the material before being finally secured in the stitch.

Mechanism of Degradation: Two Complete Passes Per Stitch

1. **DOWNWARD MOVEMENT:** the hook catches and expands the loop → the thread is drawn through the tension discs, guides, needle eye and material
2. **UPWARD MOVEMENT:** the tension regulator tightens the thread → the same section passes again through the material, needle eye and all guides in the reverse direction

Types of mechanical stress on the thread per cycle: friction against fabric fibres during downward and upward movement; friction through the needle eye; friction through metal and ceramic guides; cyclic tension with variable force (amplitude of fluctuation up to 300%); repeated bending cycles through narrow apertures (bending radius 0.1–0.3 mm); dynamic loads during abrupt changes in direction of movement.

Non-uniform movement of the thread take-up — an additional degradation factor. Lever thread take-ups create uneven, jerky thread movement: sharp accelerations and decelerations at every cycle create impact loads on the thread; amplitude of tension fluctuations reaches 300% of the nominal value; at high speeds (4,000–5,000 stitches/min) impact loads multiply sharply. Only a rotary thread take-up provides smooth movement, but it is applicable only for light materials.

Parameter	Consequence
Thread strength loss	Up to 25–35% at high speeds (4,000–5,000 stitches/min)
Hook size limitation	Practical diameter limit 35–40 mm (larger hook → even greater degradation)
Thread type restriction	Only abrasion-resistant threads; cost 20–40% higher than ordinary
Maximum reliable speed	3,500–4,500 stitches/min for most materials

> ***Conclusion on Defect No. 2:*** *The root cause — excessive thread consumption and repeated friction — CANNOT be eliminated. This is a physical limitation inherent in the very concept of the mechanism: a single section of thread MUST ALWAYS pass through the system multiple times before final securing.*

DEFECT NO. 3: CONSTRUCTIONAL COMPLEXITY AND HIGH OPERATING COSTS

The presence of a bobbin inside a rotating hook inevitably engenders mechanical complexity — a source of multiple operational problems that increase production costs.

Constructional complexity: 6–11 precision-machined moving components (hook, bobbin, bobbin case, opener, springs); phase tolerances of no more than 2° for synchronisation of all components; critical clearances and tolerances: 0.005–0.05 mm; production accuracy class: no lower than class 6 per ISO.

Lubrication requirements and their consequences. The sliding zone between the working surfaces of the hook and bobbin case requires constant lubrication. The automatic lubrication system adds 8–12 additional components. Lubrication system cost: 15–25% of the sewing head cost. Oil and filter replacement: every 200–300 hours of operation. Risk of material contamination by oil — particularly critical for delicate fabrics.

The problem of fouling in the sliding zone. Lint and thread particles continuously accumulate in narrow clearances. Contaminants block critical clearances (0.005–0.05 mm). Accumulation of dirt leads to hook seizure, sudden machine braking, risk of needle breakage and mechanism damage. Daily cleaning required during intensive production. Without cleaning: skipped stitches increase by 15–25%, mechanism temperature rises by 20–30°C.

Limited hook service life. Sliding surfaces wear at a rate of 0.001–0.005 mm per 1,000 hours. Average service life of the hook assembly: 5,000–8,000 hours with proper maintenance. Oscillating hook service life: only 3,000–5,000 hours due to impact loads.

Cost Item	Cost / Losses
Daily maintenance	15–20 minutes per machine
Lubricants	$50–100/year per machine
Maintenance downtime	5–8% of working time
Mechanic staff	1 mechanic per 15–20 machines (+8–12% to operating costs)
Defect losses due to malfunctions	3–5% of production volume
Downtime during repair	$50–100/hour per machine

DEFECT NO. 4: IMPOSSIBILITY OF GUARANTEED SKIP-FREE, BREAK-FREE AND NEEDLE-BREAK-FREE SEWING

One of the most exacting requirements of lockstitch technology is the maintenance of a microscopically small clearance between the needle and the hook point. This parameter determines the reliability of every stitch formation and is a source of constant production problems.

Critical clearance: 0.01–0.08 mm. The required clearance between the hook point and the needle scarf is 0.01–0.08 millimetres (10–80 microns). For comparison: the thickness of a human hair ≈ 70 microns. Tolerance: ±0.005 mm for high-speed machines.

Clearance Violation	Consequence
Clearance > 0.1 mm	Hook point fails to enter thread loop → SKIPPED STITCH (frequency 30–70%)
Clearance < 0.005 mm	Needle strikes hook point → NEEDLE BREAKAGE + hook point damage
Negative clearance (contact)	Instant needle breakage → hook damage → 15–30 min downtime
Incorrect adjustment	Needle tip deformation → skipped stitches + material damage

Why guaranteed sewing is impossible. This microscopic tolerance fundamentally prevents the guarantee of stable seam quality: at high speeds vibrations shift the clearance beyond tolerance; when sewing stiff and dense materials the needle deflects under lateral forces; when crossing thickened seams the needle bends; when bearings wear, needle run-out exceeds tolerance; with thermal expansion of components the clearance changes; any deterioration in needle condition immediately causes loss of loop capture.

Needle change — mandatory readjustment. Every time a different needle size is used, the clearance inevitably changes and requires obligatory readjustment of hook position. A Nm 60/8 needle (shank diameter 0.60 mm) and a Nm 130/21 needle (shank diameter 1.30 mm) have a diameter difference of 0.70 mm — the clearance changes by 0.20–0.25 mm, which exceeds the tolerance by a factor of 3–4.

Requirement	Production Reality
Operator qualification	Mechanic with at least 1 year of experience + specialist training
Mechanic training cost	$2,000–5,000 per person
Average waiting time for mechanic	15–40 minutes (depending on workload)
Adjustment time	8–15 minutes per machine
Losses on assortment change (2–3 times per day)	40–90 minutes of downtime

> ***Conclusion on Defect No. 4:*** *Lockstitch technology does NOT permit guaranteed sewing without skipped stitches, thread breaks, needle breakages and tip deformation. The microscopic clearance tolerance (up to 0.1 mm) makes the system fundamentally sensitive to any deviation from ideal operating conditions.*

DEFECT NO. 5: THREAD THICKNESS LIMITATIONS AND THE NEED FOR THE OSCILLATING HOOK

The loop exit clearance between the hook point and the bobbin case physically limits the maximum thickness of thread that can be used. Loop exit clearance for the rotating hook: 6.5–15 mm. Rotating hook: reliable operation only up to thread V-346 (Tex 400). For threads Tex 400, 600, 700 and above, the rotating hook becomes unsuitable.

For sewing with very thick threads, manufacturers are compelled to use an oscillating (pendulum) hook with a loop clearance of 20–35 mm. This gives rise to new serious limitations:

Parameter	Rotating Hook	Oscillating Hook
Maximum speed	5,000+ stitches/min	2,500–3,000 stitches/min
Vibration	Low (balanced)	High (amplitude 2–3 mm)
Operating noise	65–75 dB	80–90 dB
Stitch accuracy	±0.1 mm	±0.3 mm
Service life	8,000–10,000 hours	3,000–5,000 hours
Productivity	High (100%)	Reduced by 40–50%

Increased needle bar stroke for thick materials. When sewing materials up to 8 mm thick, the standard needle bar stroke (up to 32 mm) must be increased to 36–38 mm. Increasing the stroke leads to larger dimensions and greater mass of the mechanism, increased inertia, reduced maximum sewing speed and significant increase in machine cost.

Production consequences: factories are compelled to maintain 2–3 different types of machines; capital expenditure increases by 1.5–2 times; components are not interchangeable; reduced production flexibility.

DEFECT NO. 6: THE SPECIFIC PROBLEMS OF THE ROTATING HOOK

Type 1: Vertical axis — bobbin-case opener. In 98% of models with a vertical axis (industrial Juki, Brother, Singer), a bobbin-case opener is used: it actively rotates the bobbin case by 10–25° against the direction of hook rotation; requires perfect synchronisation with the rotation phase (accuracy ±1°); opener cam wear — 0.01–0.03 mm per 1,000 hours; phase shift >2° leads to immediate stitch defects; cost of opener replacement — 15–25% of the hook assembly cost; adds 10–15% to the cost of the sewing head.

Type 2: Horizontal axis — anti-rotation spring. In 80–90% of models with a horizontal axis (Pfaff, certain Brother models), an anti-rotation spring is used instead of an opener. Weakening of the spring by 20–30% leads to rotation of the bobbin case by 5–15° and skipped stitches. On spring breakage — full rotation of the bobbin case and machine seizure. Replacement period: 2,000–3,000 hours of operation. Frequency of service calls for this reason: 15–25%.

DEFECT NO. 7: THE CHECK SPRING OF THE TENSIONING DEVICE — A COMPULSORY ELEMENT

Lockstitch technology compels the use of a check spring in the top thread tensioning device — particularly when sewing thin materials. When sewing thin materials the needle descends rapidly and, due to the inertial lag of the thread take-up, the top thread goes slack near the needle point. The slack thread loop may catch on the needle point as it moves, leading to immediate thread breakage. The check spring maintains constant thread tension at this critical moment.

Limitations of the check spring: adds additional mechanical resistance to thread movement; increases total tension → additional load on the thread; requires precise adjustment of spring stiffness for each material type; spring wear → change in force → unstable seam quality; yet another element requiring periodic replacement and adjustment.

DEFECT NO. 8: THE NEEDLE PLATE WITH ROUND HOLE — RISK OF COLLISION

The design of the needle hole in the needle plate is a source of serious production risks. The standard round hole: its diameter must not exceed 1.5 times the needle shank diameter. The small hole diameter leads to a high probability of the needle colliding with the plate edge.

High-risk conditions: sewing stiff and dense materials — lateral forces deflect the needle; crossing thickened seams and overlaps — sharp needle bending; use of a fine needle with stiff material; incorrect needle installation (axial displacement). **Consequence:** needle breakage + tip deformation + scoring of the plate hole.

DEFECT NO. 9: COMPLEX THREAD TAKE-UP MECHANISMS — PHYSICALLY IMPOSSIBLE UNIVERSALITY

The thread take-up must perform two critical functions within 0.008–0.015 seconds (at 5,000 stitches/min): complete tightening of the top thread loop, and delivery of sufficient thread length to the needle. Smooth and uniform tightening of the top thread is physically impossible due to rigid time constraints. Industry has been compelled to develop four fundamentally different types of thread take-up:

Thread Take-Up Type	Max. Speed	Field of Application
Rotary without eye	Up to 5,000 stitches/min	Light materials only (silk, chiffon)
Crank-rocker with eye	Up to 5,000 stitches/min	Light and medium materials
Crank-slider with eye	Up to 3,500 stitches/min	Medium and heavy materials
Crank-cam with eye	Up to 2,000 stitches/min	Very heavy materials only

It is impossible to create a single universal thread take-up for materials ranging from the finest silk (0.05 mm) to thick leather (3.0 mm). This forces factories to maintain different types of machines, increasing capital expenditure by $5,000–15,000 per workstation.

DEFECT NO. 10: MANUAL BOTTOM THREAD TENSION ADJUSTMENT — DIGITAL CONTROL IS IMPOSSIBLE

Digital control of bottom thread tension is technically impossible in existing designs. The tensioning device is located on the rotating bobbin case inside the hook. It is physically impossible to connect an electronic drive to a rotating element in a confined space. Adjustment remains a manual operation only — on ALL existing machine models. This is a fundamental limitation that cannot be eliminated without abandoning the concept of the bobbin.

Factor	Production Consequence
Setting time per material	5–15 minutes by trial and error (3–5 iterations)
Losses on material change per shift	20–40 minutes per machine
Line downtime (20 machines) on assortment change	100–300 minutes (1.5–5 hours)
Productivity losses on changeover day	15–25%
Quality variation coefficient	15–25% between different operators
Operator training period	3–6 months at a cost of $800–1,500

DEFECT NO. 11: FUNDAMENTAL INCOMPATIBILITY WITH DELICATE AND ELASTIC MATERIALS

Delicate thin materials. The hook mechanism requires constant lubrication (5–15 lubrication points). Even a micro-droplet of oil (0.001 ml) creates a permanent stain on silk, satin and fine synthetics. At high speeds centrifugal force sprays micro-droplets up to 50 mm away. It is impossible to fully eliminate the risk without abandoning lubrication (without lubrication: seizure within 2–3 hours). Frequency of oil stains: 3–8% on delicate materials.

Elastic and stretch materials. The geometry of the lockstitch is fundamentally not designed for elastic applications. Seam stretch: no more than 30%, whilst the stretch of modern elastic materials is 100–200%. Stitch elasticity coefficient: 1.1–1.3 (versus 2.0–3.0 for the material). Seam failure mechanism: garment stretching → fabric extends by 50–200%, stitch threads cannot extend proportionally → critical tension → thread break → seam failure after 10–20 stretch cycles. Service life: 15–30 washes versus 100+ for special elastic stitches.

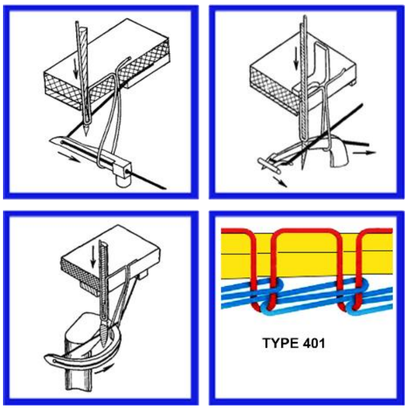
Summary Table: Why the Defects Are Fundamentally Irremediable

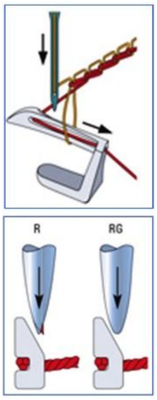
No.	Defect	Reason for Irremediability
1	Frequent bobbin replacement	The bottom thread MUST ALWAYS reside on a bobbin inside the hook; increasing the bobbin → greater inertia → reduced speed
2	Thread strength loss	The physics of friction cannot be circumvented; one section of thread MUST ALWAYS pass through the system multiple times
3	Complexity and fouling of the hook	The sliding zone MUST ALWAYS require lubrication and cleaning; seizure from fouling is inevitable
4	Skipped stitches, breaks, needle breakage	The 0.01–0.08 mm clearance is physically necessary for loop capture; any deviation → defect
5	Thread thickness and speed limitation	The loop exit clearance physically limits loop size; larger hook → slower
6	Rotating hook problems	The loop exit geometry requires either an opener or an anti-rotation spring
7	Check spring	Thread take-up inertia inevitably creates thread slack; the spring is a mandatory element
8	Collision risk with needle plate	The small hole is necessary to support the material; enlarging it → loss of function

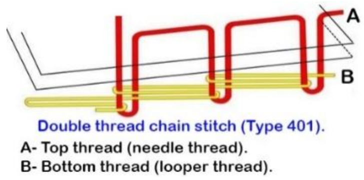
9	Thread take-up complexity	Different materials require different forces; one universal thread take-up mechanism is physically impossible
10	Manual tension adjustment	The bottom thread tensioning device rotates; a digital drive cannot physically be connected
11	Incompatibility with delicate/elastic materials	Lubrication is unavoidable → risk of staining; stitch geometry not designed for stretching

> ***Conclusion on Type 301:*** *Lockstitch type 301 technology has reached its physical and mechanical evolutionary limit. No quantity of advanced engineering solutions — whether the latest servo drives, computer control or artificial intelligence — can resolve the listed problems, since they are embedded in the very DNA of the technology at a fundamental physical level. The lockstitch dominates not because it is excellent, but because for a long time no viable alternative existed that could replace it.*

1.3. Chain Stitch (Type 401): 4 Fatal Problems







The double thread chain stitch type 401 (class 400 per ISO 4915) is formed by a single needle with top thread (thread "A") and a eye-looper with bottom thread (thread "B"). Thread interlacing occurs on the principle of "loop through loop" on the underside of the material — only the top thread passes through the thickness of the layers being joined. Theoretically, this stitch possesses a number of advantages over the lockstitch type 301: no bobbin, higher potential speed (6,000–8,000 stitches/min), increased seam elasticity (extension up to 200%). However, over 160 years of industrial use, this technology has never managed to displace the lockstitch and occupies only 15–20% of the single-needle sewn seam market.

⚠ KEY CONCLUSION

The four fundamental defects of the eye-looper technology are NOT the result of engineering errors but are the inevitable consequences of the basic physics of chain stitch formation on the "loop through loop" principle. They are irremediable without completely abandoning the concept of the eye-looper.

The Central Role of the Thread Triangle

The key element in the formation of chain stitch type 401 is the so-called thread triangle. It is formed at the moment when the needle descends for the next penetration of the material and the top thread "A" loop already rests on the looper body from the previous stitch. The thread triangle is formed by three elements: the body of the looper itself; the bottom thread "B" emerging from the looper eye; and the top thread "A" loop resting on the looper body from the previous stitch.

For the top thread to pass through the bottom thread loop and form a proper stitch, the needle must reliably pass through this thread triangle. It is precisely this requirement that generates the entire chain of irremediable problems of this technology.

The Two-Stage Process of Top Thread Loop Tightening

Stage 1: Preliminary tightening by the needle (downward movement). After the top thread loop is shed from the looper body, the only way to begin tightening it is to use the downward movement of the needle. The thread take-up deliberately delivers to the needle an insufficient length of top thread — less than is required for the needle to reach its lowest position. Due to this deficit, the needle is compelled to draw the top thread out of the previous stitch loop, thereby performing preliminary tightening. The top thread consumed by the needle during descent equals twice the distance from the upper surface of the material to the needle eye at its lowest position.

For successful completion of Stage 1, a strict condition is required: the tension created by the top thread tensioning device must be GREATER than the total friction of the top thread against the material at three points (current penetration, previous penetration, upper surface). **Irremediable limitation:** friction of the top thread against the material at the PREVIOUS penetration point is technically impossible to reduce — there is no needle at that point. On dense materials (leather, denim, technical fabrics), the total friction exceeds the tensioning device force (0.8–1.2 N) — the needle begins to draw thread from the cone rather than from the previous stitch loop. The loop remains untightened.

Stage 2: Final tightening by the looper body. The looper moves in a direction opposite to the material feed, simultaneously tightening both the top thread "A" loop and the bottom thread "B" loop. **Key limitation:** if at Stage 1 the needle failed to perform adequate preliminary tightening and left excess top thread loop length, the looper cannot compensate for this excess. The looper stroke is fixed (8–15 mm) and it is impossible to take up the excess thread in full.

Stage	Maximum Achievable Tightening Force
Stage 1: preliminary tightening by the needle	0.5–0.7 N
Stage 2: final tightening by the looper	up to 1.2 N total
Tightening force of lockstitch type 301 (for comparison)	2–3 N (thread take-up with direct drive)
Chain stitch tightening force deficit	**30–40% lower than type 301 stitch**

The Compulsory Use of a Special Two-Groove Needle

Chain stitch type 401 technology with an eye-looper compels the use of a special needle with two longitudinal grooves instead of the standard needle with one long groove. This compulsory constructional decision is itself a shortcoming, as it reduces the mechanical reliability of the needle. The second longitudinal groove reduces friction of the top thread at the current penetration point, but diminishes the moment of inertia of the needle shank cross-section. According to the technical report of Organ Needle (2023), a needle with two longitudinal grooves loses 35–40% of its resistance to lateral bending compared with the standard needle.

Parameter	Standard Needle (1 groove)	Special Needle (2 grooves)
Top thread friction at penetration point	High	Reduced (objective achieved)
Cross-section moment of inertia I	Base	Reduced by 35–40%
Resistance to lateral bending	Base	Reduced by 35–40%
Deflection δ under equal load	Base	Increased by 55–65%
Risk of collision with looper	Moderate	Significantly higher

The vicious circle of constructional compromise: the second groove reduces friction at the penetration point (which helps Stage 1), but simultaneously increases needle deflection when sewing dense materials. Increased deflection raises the probability of needle collision with the looper body. Thus, a measure intended to improve tightening quality simultaneously worsens the reliability of passage through the thread triangle.

FLAW NO. 1: TWO-STAGE TIGHTENING DOES NOT PROVIDE FIRM CLAMPING OF MATERIALS

This is the most significant systemic shortcoming, fundamentally limiting the field of application of chain stitch type 401. In lockstitch type 301, both threads pass through the thickness of the material and interlace precisely in the middle. A thread take-up with direct drive creates a symmetrical tightening force of 2–3 N, firmly pressing the material layers together. In chain stitch type 401, the threads interlace on the underside of the material. Only the top thread "A" passes through the thickness. The only way to achieve firm clamping is to tighten the top thread loop as forcefully as possible. And it is precisely this that proves impossible.

Consequences of weak tightening: clamping pressure 30–40% lower than lockstitch type 301; total tightening force does not exceed 1.2 N against 2–3 N for type 301 stitch; the top thread loop remains partially untightened on the underside of the material; seam thickness on the reverse side increases by 1–2 mm; seam elasticity on stretching is increased by 20–30%; seam wear resistance is 1.5–2 times lower; technology unsuitable for joining material combinations: textile-leather, textile-plastic, textile-zip fastener.

> ***Conclusion:*** *Friction of the top thread against the material at the previous penetration point cannot be reduced by constructional means. The looper stroke is fixed — it is impossible to compensate for excess thread at Stage 2. This is a physical limitation deriving from the principle of "only the top thread through the material."*

FLAW NO. 2: THE THREAD TRIANGLE PROBLEM AND MINIMUM STITCH LENGTH LIMITATION

The area of the thread triangle is linearly dependent on stitch length: $S \propto t$. As stitch length decreases, the triangle diminishes proportionally, and at some critical value the triangle area becomes less than the cross-sectional area of the needle — reliable needle passage becomes physically impossible.

Looper Type	Minimum Reliable Stitch Length	Triangle Area at t=3–4 mm
Complex spatial movement	$t \geq 1.2$ mm	15–20 mm²
Vertical with expander	$t \geq 1.0$ mm	12–16 mm²
Arc (oscillating) movement	$t \geq 1.5$ mm	6–9 mm²

Why this is critical for seam tacking. Optimal tacking of a chain seam end using condensation stitches requires a stitch length of 0.5 mm or less. Only at such a length is friction between the threads inside the tacking stitches sufficiently great to reliably resist a load of 40–50 N. However, when $t < 1$ mm, the triangle area falls below 2–3 mm² — less than the cross-sectional area of the needle (diameter 0.8–1.0 mm). The minimum passability condition is violated: the needle cannot pass without collision or skip. The probability of a skipped stitch at $t < 1$ mm is 30–70%.

Tacking Parameter	Ideal	Reality of Eye-Looper Technology
Required tacking stitch length	0.5 mm	Technically impossible (skips >30–70%)
Actually achievable tacking length	—	≥ 1.0–1.5 mm → insufficient friction
Tacking zone length at 10 stitches	5 mm	10–15 mm
Holding force against unravelling	40–50 N	5–12 N
Tacking reliability on dense materials	100%	40–60%

FLAW NO. 3: NEEDLE–LOOPER COLLISION — AN IRREMEDIABLE HAZARD

For the needle to pass reliably through the thread triangle, it must travel as close as possible to the eye-looper body. This is a direct reliability requirement — the closer to the eye-looper body, the lower the probability of straying outside the triangle. However, it is precisely this requirement that makes needle-looper collision a constant threat.

Clearance between needle and eye-looper body: 0.05–0.15 mm (type 1), 0.1–0.3 mm (type 2), less than 0.1 mm (type 3). Needle deflection when sewing dense materials is calculated by the Euler-Bernoulli formula: $\delta = F \times L^3 / (3 \times E \times I)$. On dense materials (denim, leather, technical fabrics) the lateral force F rises significantly, and deflection δ reaches 0.2–0.3 mm — which exceeds the working clearance by a factor of 1.5–3. Collision is inevitable.

High-risk conditions: sewing stiff and dense materials; crossing thickened seams and overlaps; use of the special two-groove needle (35–40% less resistant to bending); use of fine needles on dense materials; any wear of bearings or guides.

Consequence of Collision	Production Effect
Needle breakage	Needle consumption 2–3 times above normal
Needle tip deformation	Material damage, deterioration in penetration quality
Looper eye damage	Accelerated wear, disruption of triangle geometry
Mechanism synchronisation disruption	Readjustment 30–60 min, line downtime
Breakage-related downtime	1–3 times per shift when working with dense materials
Cost increase	+25–40% on maintenance and consumables

IRRESOLVABLE CONTRADICTION:

Reliable passage through the thread triangle requires maximum proximity of the needle to the eye-looper body. Proximity of the needle to the eye-looper increases the risk of their collision. Distancing the needle from the eye-looper reduces the reliability of passage through the triangle. This contradiction cannot be resolved by constructional means within the framework of this technology.

FLAW NO. 4: INHERENT SUSCEPTIBILITY OF THE SEAM TO UNRAVELLING

The "loop through loop" principle — the foundation of any chain stitch — is simultaneously the source of its principal vulnerability. Each stitch of the chain seam is held only by the next stitch. When any link of the chain breaks, the seam begins to unravel from the point of breakage.

Two sources of unravelling: open (unsecured) seam end — probability of spontaneous unravelling 60–80%; a skipped stitch at any point in the seam — the seam unravels from the point of the skip in the reverse direction. Unlike the lockstitch type 301, where a skipped stitch merely degrades seam quality at that point, in the chain stitch any skip breaks the continuity of the chain. The result: progressive seam unravelling from the skip point backwards to the beginning.

Two seam-end tacking methods and their shortcomings. Tacking a chain seam by "back tacking," as with lockstitch type 301, is completely ineffective — the seam will continue to unravel. *Method 1: Thread chain beyond the material edge* — continuing to sew beyond the material edge creates a thread chain that mechanically prevents unravelling, but it degrades appearance, increases thread consumption, accelerates wear of the presser foot and feed dog teeth, and is applicable only if the seam end is located at the material edge. *Method 2: Condensation stitches of minimum length* — reducing stitch length increases friction between the threads, but at $t < 1$ mm the probability of a skipped stitch rises to 30–70%, and a skip during tacking immediately triggers unravelling of the entire seam.

Condition	Consequence for Chain Seam Tacking
Abrasion zones (elbows, knees)	Tack fails under a load of 5–8 N

Repeated stretching	Unravelling after 10–20 cycles
High seam load	Failure at load >5–12 N (versus 40–50 N norm)
Skipped stitch in tacking zone	Immediate unravelling of the entire seam
Product service life	1.5–2 times shorter than with type 301 stitch

Three Types of Eye-Looper — Three Unsuccessful Attempts to Circumvent Limitations

Three types of eye-looper represent three attempts to constructionally circumvent the common systemic limitations of the technology. Each has its own specific problems in addition to the four fundamental flaws.

Looper Type	Specific Problems	Maximum Speed
Complex spatial movement	Up to 20 kinematic elements; fatigue failures at vibration >10g; +50–70% to mechanism cost; instability at frequency >5,000 stitches/min	5,000–6,000 stitches/min
Vertical with expander	Rapid expander wear (service life 500–1,000 h); synchronisation accuracy ±1° phase; thread slippage after 500 h; clearance 0.1–0.3 mm	4,000–5,000 stitches/min
Arc (oscillating) movement	Smallest triangle area (6–9 mm²); minimum reliable stitch length 1.5 mm; worst tightening of the three types	3,000–4,000 stitches/min

All three types of eye-looper share the same four fundamental flaws. The differences between them lie only in the degree to which these manifest and in the ratio of "maximum speed — construction cost — tacking quality." None of the three types eliminates the systemic limitations of the technology.

Why, over 160 years, the technology failed to replace the lockstitch: *The combination of four fundamental flaws makes chain stitch type 401, formed using an eye-looper, unsuitable for 80–85% of industrial applications where the lockstitch dominates. Firm material clamping — not provided (tightening force 30–40% below normal). Reliable seam-end tacking — not provided (minimum tacking stitch ≥1 mm, holding force 5–12 N instead of 40–50 N). Guaranteed skip-free sewing — not provided (skip probability 5–15%). Sewing dense and stiff materials — limited (risk of needle-looper collision). Sewing through thickened seams — limited (peak needle deflection > working clearance). The only escape from this dead end is a transition to a fundamentally different stitch-formation architecture — free from the eye-looper and its associated physical limitations.*

1.4. The Graveyard of Robotisation: Why $220 Million Yielded No Result

From 2010 to 2025, the industry tried every conceivable approach: high-speed machine vision (SoftWear), adaptive robotic manipulators (Siemens/Adidas), even temporary "stiffening" of fabric with polymers (Sewbo). All projects collided with the same wall: the mechanical architecture of the sewing machine is not designed for autonomous operation. A robot cannot replace a human if the machine demands manual bobbin replacement every 15 minutes or readjustment of a 0.05 mm clearance.

The gap is evident: robots are ready, AI is ready, but the tool — the sewing machine — is obsolete. It is precisely this gap that ZARIF 2025 technology addresses, having been designed from the outset as a robot-native device.

CHAPTER 2. THE FORGOTTEN GENIUS AND THE BIRTH OF AN IDEA

How the Rotary Looper of 1857 Waited 137 Years to Change the World

2.1. 1857: James Gibbs and the Machine Ahead of Its Time

Long before a question that would overturn an entire industry was asked in Tashkent, the answer already existed — in the form of a forgotten patent from the mid-19th century. On 2 June 1857, American inventor James Edward Allen Gibbs received US Patent No. 17,427 for the world's first single-thread chain stitch sewing machine with a rotary looper.

Unlike the oscillating loopers then prevailing — which jerked back and forth, creating vibration and a broken rhythm — the Gibbs mechanism rotated continuously. It was an engineering masterpiece: a monolithic component without internal moving parts, without a bobbin, without an eye. The looper caught the top thread loop, expanded it, rotated it through 180° using a special "tail" section, and released it through the rear beak — all in a single smooth revolution.

Machines made by the Willcox & Gibbs Sewing Machine Company became legendary. They were celebrated for their exceptional reliability: many examples produced in the 1860s–1880s remain in working order to this day — living testimony to the brilliance of the design.

2.2. Anatomy of the Rotary Looper: Four Zones of Excellence

Why did the Gibbs looper prove so successful? Because a single component combined four functional zones, each performing a strictly defined task:

Zone	Component	Function
1. Head	Front and rear beaks	Catching the top thread loop and releasing it after rotation
2. Tail section	Special inclined surface	Rotating the loop through 180° — geometric manipulation of thread orientation
3. Heel	Section between rear beak and tail	Holding the loop in its maximally extended state for guaranteed entry of the next loop
4. Shank	Cylindrical base	Rigid attachment to shaft, transmission of rotational motion

This principle — continuous rotation instead of oscillation — was so elegant that it seemed it could be applied immediately to a two-thread stitch. But here engineers the world over ran into what appeared to be an insurmountable wall.

2.3. The 137-Year Barrier: Why No One Could Adapt Gibbs' Idea

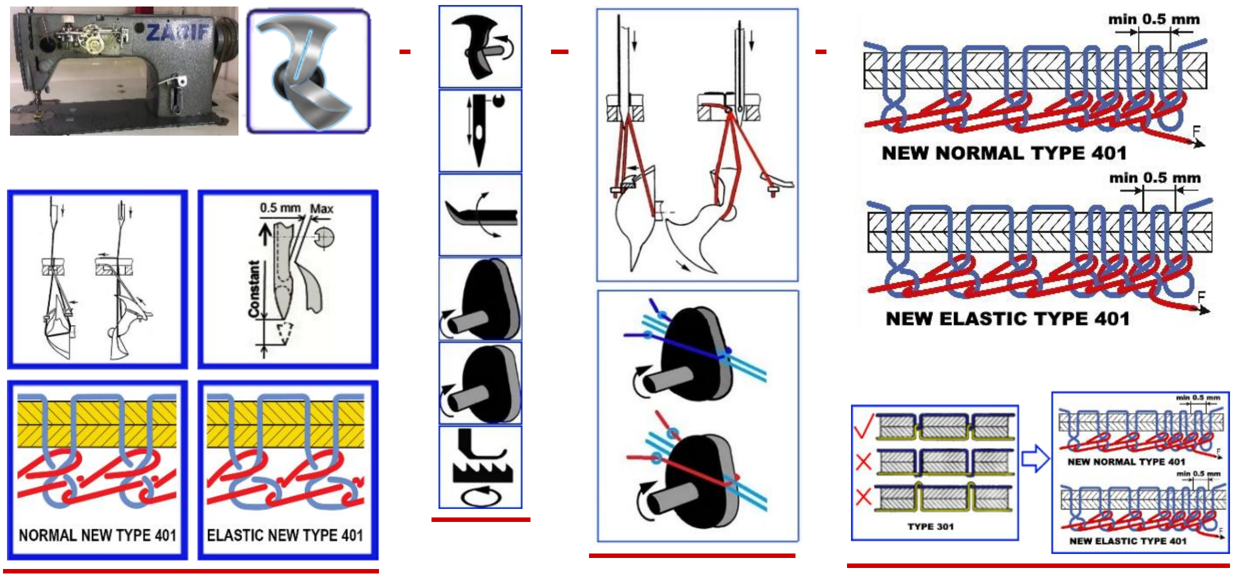
The single-thread chain stitch uses only one thread. It naturally forms a loop when the needle rises from the material. The looper simply manipulates this loop. But the double thread chain stitch requires a second, bottom thread. In traditional machines it was passed through the looper eye. And Gibbs' rotary looper has no eye. It cannot carry thread with it. It can only catch, expand and turn already-formed loops.

This incompatibility was accepted as axiomatic. From 1857 to 1994 — 137 years — not a single engineer in the world was able to circumvent this limitation. Thousands of patents were granted for improvements to eye-loopers: oscillating, spatial, with expanders. But no one dared remove the eye entirely. Everyone believed that the rotary looper and the double thread chain stitch were incompatible.

THE HISTORICAL PARADOX:

Tens of thousands of engineers, hundreds of companies with enormous budgets, billions of dollars in investment — and not a single successful attempt. The problem was considered insoluble by definition.

2.4. Tashkent, 1994: The Question Nobody Had Asked



In 1994, Zarif Tadjibaev, a young postgraduate student at the Tashkent Institute of Textile and Light Industry, was working on his dissertation on new methods of chain stitch formation. He studied all existing designs thoroughly and saw what had eluded his predecessors for decades: the root of all problems is the eye-looper. And then he posed the question that would transform his life and soon overturn an entire industry:

"Is it possible to create a double thread chain stitch type 401 without an eye-looper through which the bottom thread passes?"

This was not a technical question. It was a challenge to 137 years of dogma.

2.5. The Revolutionary Insight: Separate the Functions

The answer that Zarif Tadjibaev found was as brilliant as it was simple. If the rotary looper cannot carry the bottom thread — let someone else carry it. The functions must be separated:

- The looper remains the "loop manipulator" — it catches, expands, turns and interlaces already-formed loops of both threads.
- The **bottom thread pusher** (a completely new element) delivers the straight branch of the bottom thread precisely into the looper's zone of action at a strictly defined moment of the cycle.

The bottom thread never passes through the body of the looper. The looper simply has no eye. This fundamental change eliminates the structural weakness that had generated all the problems of traditional machines.

2.6. Two Rotations Instead of One: The Key to the Double Thread Chain Stitch

But separation of functions alone was insufficient. In the Gibbs machine, the looper made one rotation per needle cycle, handling only one thread. The double thread stitch requires more actions. The solution came after careful kinematic analysis: **the looper must make two full rotations per needle cycle.**

Rotation	What Happens
First	Catching the top thread loop, expanding it, rotating through 180°, receiving the bottom thread from the pusher, tightening the top thread loop, first interlacing

Second	Forming the bottom thread loop after tightening of the top thread loop, expanding it, rotating through 180°, second interlacing with a new top thread loop, final tightening

This was an architectural solution without precedent. No engineer before Zarif Tadjibaev had conceived of making a rotary looper perform two rotations per cycle. It was precisely this that made it possible to eliminate entirely the "thread triangle," the critically small clearances, the needle's involvement in loop tightening, and all the other problems that had dogged the traditional double thread chain stitch technology with an eye-looper.

2.7. US Patent No. 6,095,069: World Recognition of Priority

Recognising the scale of the discovery, Zarif Tadjibaev immediately began the process of international patenting:

- **1994** — Application filed at the Patent Office of Uzbekistan.
- **1995** — International application PCT/UZ95/00001 registered at WIPO in Geneva.
- **1996** — Positive international search report from WIPO received: world novelty, inventive step and industrial applicability confirmed.
- **2000** — The USPTO grants **US Patent No. 6,095,069** — official recognition that a problem deemed insoluble for 137 years has at last been solved.

✓ DOCUMENTED PROOF:

US Patent No. 6,095,069 is the first patent in history for a method of forming a double thread chain stitch using a rotary looper. This is not theory. It is legally protected priority.

2.8. From Patent to Prototype: 1997 and the First Working Model

A patent describes a principle, but only a working prototype proves its viability. In 1997, Zarif Tadjibaev created the first operational model, taking as a basis an industrial lockstitch machine of the 1022 class from the Orsha plant (Belarus) and completely rebuilding its "heart":

- **Removed:** the entire hook mechanism, the bobbin case, the lever thread take-up, the automatic bobbin-winding system.
- **Installed:** a rotary looper of original design, a bottom thread pusher, two dual-disc cam thread take-ups (for top and bottom thread), and a new lower-shaft system with a 2:1 transmission ratio.

This prototype was far from perfect: straight-cut gears created noise, cast-iron bearings required manual lubrication, and exposed mechanisms did not meet safety standards. But it sewed. And in doing so, it proved that the idea born in 1994 works in metal. All that was required thereafter was 28 years of persistent effort to transform a rough prototype into the ideal ZARIF 2025 technology.

CHAPTER 3. THE ANATOMY OF REVOLUTION: ZARIF 2025

Dual-Rotation Kinematics, 13 Phases of the Perfect Stitch, and the Birth of a New Type of Chain Seam

✓ THIS ENTIRE CHAPTER IS BASED ON PROVEN FACTS

The described mechanism is realised in metal and confirmed by the ZARIF 2025 prototype. Video demonstration is available on the YouTube channel @ZarifTadjibaev.

3.1. Two Rotations That Changed Everything

The heart of ZARIF 2025 technology is the rotary looper, making two full rotations per single needle movement cycle. This is the key architectural solution, without precedent in the history of sewing machine engineering. It is precisely this that made it possible to separate the functions of loop manipulation and bottom thread delivery, completely eliminating the eye-looper.

Unlike the traditional chain stitch, where the eye-looper performs complex spatial movement with critical clearances in hundredths of a millimetre, the ZARIF 2025 rotary looper operates in the gentle regime of continuous rotation. This radically reduces vibrations, eliminates "dead zones" and makes the stitch-formation process predictable.

Two rotations are required not for speed — they are required for the sequential processing of two threads. The first rotation catches the top thread loop, expands it, rotates it through 180° and introduces into it the bottom thread delivered by the pusher. The second rotation forms the bottom thread loop, expands it, rotates it and completes interlacing with a new top thread loop. This sequence guarantees perfect "loop through loop" interlacing without the geometric constraints of the "thread triangle."

3.2. Thirteen Phases in the Birth of the Perfect Stitch

The formation of a single stitch in ZARIF 2025 technology is a precisely choreographed sequence of 13 consecutive phases, each synchronised to within fractions of a millisecond. Let us examine them in detail.

First Rotation: Catching the Top Thread and Introducing the Bottom Thread

Phase	Action	Key Advantage over Traditional Technology
1. Catching the top thread loop	The front beak of the looper enters the loop formed by the needle as it rises from its lowest position.	The clearance between needle and beak can reach 0.5 mm — 5–10 times greater than with a traditional eye-looper. The catch is stable even when the needle deflects.
2. Expanding the loop	The looper continues rotating, expanding the caught top thread loop to its maximum size.	Smooth expansion on the smooth looper body without sharp angles eliminates thread damage.
3. Rotating the loop through 180°	The tail section of the looper with its special inclined surface rotates the top thread loop through 180 degrees.	It is precisely this rotation that creates the unique appearance of the reverse side — flat and neat, like the lockstitch. Traditional chain stitch has no such rotation, which is why the reverse side looks like a voluminous "chain."
4. Delivering the bottom thread	The bottom thread pusher delivers the straight branch of the bottom thread precisely into the looper's zone of action.	The bottom thread does not pass through an eye — it is simply "offered" to the looper. Friction is minimal; risk of breakage excluded.
5. Catching the bottom thread by the looper body	The looper body catches the delivered branch of the bottom thread. The branch must lie within the circular trajectory of the front beak.	The pusher guarantees precise positioning. Probability of failure to catch is reduced to zero by the special geometry of the pusher and synchronisation.
6. First interlacing	The front beak of the looper enters the top thread loop held taut on the heel, and inserts the caught bottom thread branch into the expanded top thread loop.	Interlacing occurs with the loop at maximum expansion — no "thread triangle," no risk of a skip.

Second Rotation: Forming the Bottom Thread Loop and Final Interlacing

Phase	Action	Key Advantage

7. Beginning of top thread loop tightening	The cam top thread take-up begins smoothly shortening the top thread loop.	Unlike the jerky lever thread take-ups, the dual-disc cam mechanism provides absolutely smooth tightening without peak loads. This is the principal reason for the absence of breaks.
8. Forming the bottom thread loop	As the top thread loop tightens, the bottom thread branch that has passed through it begins to bend and forms its own loop.	The process proceeds naturally, without forced pulling by the needle. The bottom thread experiences no overloading.
9. Expanding the bottom thread loop	The looper, continuing its second rotation, expands the just-formed bottom thread loop.	The looper heel holds the loop in its maximally expanded state for guaranteed entry of the front beak.
10. Rotating the bottom thread loop through 180°	The tail section again performs the rotation — this time of the bottom thread loop.	Both loops are turned equally. This creates the unique flat reverse side — the hallmark of the new ZARIF stitch.
11. Preparation for the next cycle	The needle makes a new penetration, passing in front of the bottom thread loop, and on its rise forms a new top thread loop.	Thanks to special needle-positioning know-how, the needle always ends up in front of the bottom thread loop, even at a minimum stitch length of 0.5 mm.
12. Second interlacing	The front beak catches the new top thread loop and enters the bottom thread loop held on the heel. The looper inserts the top thread loop into the bottom thread loop.	Again — interlacing at maximum loop expansion, with no geometric constraints.
13. Final tightening	The bottom thread take-up tightens the bottom thread loop. The cycle is complete.	Smooth tightening, identical to the top thread. Both threads tightened with equal force, ensuring perfect stitch balance.

KEY DISTINCTION FROM TRADITIONAL TECHNOLOGY:

The needle NEVER participates in loop tightening. It only penetrates the material and carries the thread. All work of tightening and interlacing is performed by the looper and the two independent cam thread take-ups. It is precisely this that frees the technology from the limitations that have dogged the chain stitch for 166 years.

3.3. The 180 Degrees That Transform Aesthetics

One of the most striking and commercially significant achievements of ZARIF 2025 technology is the rotation of both loops through 180°, which radically transforms the appearance of the reverse side of the chain stitch. This is not merely a cosmetic improvement — it is a strategic advantage, opening the doors of markets to the chain stitch where previously the lockstitch held undivided sway.

Aspect	Traditional Chain Stitch Type 401	New Chain Stitch ZARIF 2025
Reverse side appearance	Voluminous, loose "chain," noticeably protruding above the material surface	Flat, tight, attractive appearance visually similar to the lockstitch
Seam profile	Thickening on reverse side (up to 1–2 mm), increases product bulk	Minimal profile; seam lies almost flush with the material surface
Susceptibility to snagging	High — the protruding chain catches easily on objects, fasteners, equipment	Low — the flat seam has minimal contact area with external objects
Abrasion resistance	Lower — the protruding chain is subject to direct mechanical wear	Higher — the flat seam is protected from direct abrasion; significantly increased service life
Quality perception	Consumers associate the voluminous chain with budget production	The flat professional appearance is associated with premium quality, characteristic of the lockstitch

Thanks to this aesthetic revolution, ZARIF 2025 becomes the first chain stitch technology suitable for premium products: suits, coats, leather accessories, reversible clothing where the reverse side may be visible. Manufacturers no longer need to choose between the speed/elasticity of the chain stitch and the appearance of the lockstitch — ZARIF 2025 delivers both.

3.4. Why 28 Years of Experiments Is Not a Shortcoming, but Proof

The prototype on which the technology was developed was created as far back as 1997. By the time of the final tests of 2025, its mechanisms had passed through thousands of hours of operation, hundreds of modifications and inevitable wear. Play appeared, clearances increased, bearings lost their original precision. For any traditional machine, such a condition would mean only one thing — write-off.

But the ZARIF 2025 prototype continued to sew. And not merely to sew — to demonstrate results unattainable by new machines of traditional types. This paradox is the best proof of the correctness of the underlying principles. Wide tolerances (clearance up to 0.5 mm), the absence of critical dependence on positioning accuracy and smooth thread tightening make the ZARIF technology remarkably resistant to wear. A machine based on this technology will operate at high quality even when traditional machines have long since required a major overhaul.

✓ PROVEN BY THE PROTOTYPE:

The 13-phase stitch-formation cycle is realised in metal. The 180° loop rotation is confirmed visually — the reverse side of the seam looks like a lockstitch. Video demonstration available on the YouTube channel @ZarifTadjibaev. The stitch-formation zone beneath the needle plate is deliberately not shown in close-up — this is part of the intellectual property protection strategy until patenting is complete.

CHAPTER 4. TEN BREAKTHROUGHS TRANSFORMING THE INDUSTRY

From Zero Needle Collision Risk to the Guarantee of Defect-Free Sewing

✓ ALL TEN BREAKTHROUGHS ARE CONFIRMED BY THE ZARIF 2025 PROTOTYPE

Each of the achievements described below is realised in metal and proved by years of testing on a worn prototype. This is not theory — it is a working technology.

Twenty-eight years of systematic experimentation, thousands of hours of testing and hundreds of engineering iterations have produced a technology that does not merely improve upon existing methods but fundamentally redefines the very possibility of industrial sewing. Below are the ten key breakthroughs of ZARIF 2025, each of which in isolation would constitute a revolution, and together they form a new paradigm.

Breakthrough No. 1: Zero Probability of Needle–Looper Collision

In traditional chain stitch machines, the clearance between the needle and the looper body is 0.05–0.15 mm. When penetrating dense or multi-layer materials, the needle inevitably bends by 0.2–0.3 mm — and strikes the looper. The result: needle breakage, tip deformation, calling a mechanic.

In ZARIF 2025, collision is **constructionally impossible**. The geometry of the rotary looper is such that even at extreme needle deflection (up to 1–2 mm), its trajectory does not intersect with the looper body. The looper is simply not located in the zone where the needle can deflect. This is achieved because the looper manipulates loops rather than "squeezing past" the needle.

Practical effect: sewing ultra-dense materials (up to 8 mm), radical increase in needle service life, complete elimination of associated downtime.

Breakthrough No. 2: Elimination of the Needle's Role in Loop Tightening

In the traditional chain stitch, the needle is compelled to perform preliminary tightening of the top thread loop from the previous stitch, overcoming friction at the previous penetration point. This critically loads the needle, requires a special two-groove needle and creates enormous resistance on dense materials.

In ZARIF 2025, the needle is **completely released from tightening**. This function is performed by two independent cam thread take-ups — for the top and bottom thread. The needle only penetrates the material and carries the thread.

Consequences: the possibility of using standard DPx5 needles (with one groove), which are 35–40% stronger than special chain stitch needles; sewing any combination of materials (textile+leather, textile+plastic); the practical disappearance of thread breaks due to needle overloading.

Breakthrough No. 3: Micro-Stitches of 0.5 mm on Materials up to 8 mm Thick

The traditional chain stitch cannot work with a stitch length below ~1.0 mm: the "thread triangle" collapses, there is nowhere for the needle to enter, and the probability of a skipped stitch is 30–70%. This made reliable seam-end tacking impossible.

ZARIF 2025 **stably forms stitches of 0.5 mm** on any material, including 8 mm packages. This is achieved thanks to a special needle-positioning know-how that guarantees the needle always ends up in front of the bottom thread loop even at minimum feed.

Practical effect: for the first time in history — compact chain seam end tacks (6–10 stitches of 0.5 mm), comparable in reliability to lockstitch tacks, but without reversal and thickening.

Breakthrough No. 4: Sewing Without Adjustment on Material Change (0.1–8 mm)

In a traditional factory, changing the material means 5–15 minutes of tension adjustment, presser foot pressure, feed dog height, and often calling a mechanic to readjust the looper clearance. This consumes up to 25% of working time.

ZARIF 2025 sews everything — from the finest silk to 8 mm leather — **on a single setting**. The operator sets "normal" thread tension once, and the machine is ready to work with any material within the range. No mechanical adjustments are required when changing fabric.

Economic effect: for a factory of 100 machines — saving of 50–100 hours of retooling per shift.

Breakthrough No. 5: Slot Instead of Hole in the Needle Plate

The traditional needle plate in lockstitch machines (except needle-feed machines) has a round hole, whose edge acts as a "guillotine" for a deflected needle. When the needle deflects by as little as 0.5 mm, it strikes the edge and breaks.

ZARIF 2025 uses a **technological slot** that allows the needle to deflect by 1–2 mm in the feed direction without contact with the plate. Even at extreme deflection on ultra-dense materials, the needle remains within the slot.

Synergy of two protections: no collision with the looper (Breakthrough No. 1) + no collision with the plate (Breakthrough No. 5). ZARIF 2025 operates with practically no needle breakages or tip deformation.

Breakthrough No. 6: Dual-Disc Cam Thread Take-Ups

Traditional lever thread take-ups in lockstitch machines operate with jerks: rapid acceleration, instantaneous stopping, reversal. This creates peak loads on the thread, provokes breaks and limits maximum speed.

ZARIF 2025 uses **rotating dual-disc cam thread take-ups** — one each for the top and bottom thread. Two profiled discs rotate synchronously with the main shaft, and the thread slides smoothly over their profile, without a single jerk.

Practical result: thread breaks practically disappear. When using quality thread without knots, the machine operates for hours without a single break.

Breakthrough No. 7: Adjustable Loop Tightening Force Without Changing Tension

Different materials require different tightening forces: for denim and leather — maximum layer compression, for knitwear — elasticity, for silk — delicacy. In traditional machines, this requires readjusting thread tension, which disrupts stitch balance.

ZARIF 2025 has **three tightening modes**, switched by a single control, without changing thread tension from the cones:

- **Strong tightening** — denim, leather, workwear, upholstery. Firm clamping, like the lockstitch.
- **Moderate tightening** — knitwear, sportswear, underwear. Elastic seam.
- **Light tightening** — silk, chiffon, organza. Perfectly smooth seam without gathering.

This know-how allows a single machine to replace three specialised ones.

Breakthrough No. 8: Clearance of 0.5 mm Between Needle and Front Looper Beak

The traditional clearance is 0.05–0.15 mm. When changing from needle No. 70 to No. 110, a mechanic must be called and 8–15 minutes of precise adjustment carried out.

In ZARIF 2025, the clearance is **increased to 0.5 mm** without losing reliability of loop capture. This means the operator can change the needle from the finest (No. 60/8) to the thickest (No. 130/21) in 15–30 seconds, simply unscrewing and tightening a single screw. No looper adjustment is required.

Revolutionary consequence: elimination of dependence on skilled mechanics for the routine operation of needle change.

Breakthrough No. 9: Sewing Materials up to 8 mm with a Needle Bar Stroke of 32 mm

Traditional machines for sewing materials 8 mm thick require a needle bar stroke of 36–48 mm. The longer the stroke, the greater the vibration, inertia, machine dimensions and the lower the maximum speed.

ZARIF 2025 sews 8 mm **with a needle bar stroke of only 32 mm** — the minimum in its class. This is achieved because the needle does not participate in tightening and does not draw thread from the previous stitch, so it requires no additional stroke for "pull-through."

Advantages: compact head, reduced vibration, capability of operating at high speeds, ideal compatibility with robotised systems.

Breakthrough No. 10: Guaranteed Defect-Free Sewing

The sum of all nine preceding breakthroughs. When three basic conditions are observed — quality thread without knots, correct needle size for the material, normal thread tension — ZARIF 2025 technology guarantees:

- Zero skipped stitches on any materials and speeds up to 5,000 stitches/min.
- Practical absence of thread breaks (only with an obvious thread defect).
- Zero needle breakages from collision with looper or plate.
- Zero needle tip deformation from impact loads.

This guarantee is unprecedented. No traditional sewing technology can offer anything comparable. It is precisely this that makes ZARIF 2025 the world's first technology suitable for fully autonomous, unmanned production, where a defect-related stoppage is unacceptable.

COLLECTIVE EFFECT OF TEN BREAKTHROUGHS:

ZARIF 2025 is not a sum of improvements but a new systemic wholeness. Each breakthrough reinforces the others, creating a multiplicative effect of reliability, versatility and economic efficiency.

CHAPTER 5. THE PROTOTYPE UNDER EXTREME TESTING

28 Years of Experiments and Proof of Excellence

✓ ALL RESULTS IN THIS CHAPTER ARE CONFIRMED BY THE ZARIF 2025 PROTOTYPE

The tests described below were conducted on a real prototype in a worn state after 28 years of experiments. Video recording is available on the YouTube channel @ZarifTadjibaev.

5.1. The Testing Philosophy: Why Wear Became the Best Proof

Ordinarily, demonstration models are prepared for shows with particular thoroughness: every unit is adjusted to the micron, lubrication is refreshed, clearances are set with maximum precision. Such a demonstration proves only that, under ideal conditions, the machine performs acceptably. But ideal conditions are unattainable in real production.

Zarif Tadjibaev deliberately took a different path. For the final tests of 2025, a prototype created back in 1997 was used — one that had passed through thousands of hours of experiments, hundreds of modifications and decades of operation. Its condition was far from ideal: perceptible play in mechanisms, worn bearings, enlarged clearances, no overhaul since its creation. If such a machine can sew perfectly — the technology genuinely works.

5.2. Test Conditions: Honesty above All

Parameter	Value
Thread	Chinese manufacture, No. 40 S/2 (100% polyester), standard quality, without special pre-selection
Needle	Standard single-groove Nm.110/18 System 134(R) DPx5 (Groz-Beckert)
Stitch lengths	3 mm and 5 mm — for main seams; 0.5 mm — for tacking stitches
Tension adjustments	Set once at the beginning of the demonstration and not changed throughout all tests
Machine condition	Worn (play, enlarged clearances after 28 years of experiments, no overhaul)

KEY CONDITION:

In none of the 10 demonstration series were any adjustments made to thread tension, needle position or looper position. All sewing was performed at the same settings established at the very beginning.

5.3. Ten Tests That No Traditional Machine Could Pass

Test No. 1: Medium-Weight Denim — 2, 6 and 14 Layers

The machine sequentially sewed 2, 6 and 14 layers of denim fabric without any adjustments. Stitch quality remained consistent across all thicknesses; no skips or breaks were recorded. Even on 14 layers, the stitch line was even and tight.

Test No. 2: Heavy Wool Cloth — 3 and 5 Layers

Three and five layers of dense wool cloth were sewn without a single issue. Thread did not break; loops formed cleanly; the pile created no interference with the stitch-formation process.

Test No. 3: Heavy Denim — 2 Layers with Two Thickened Seams (up to 10 Layers in Seam Zones)

The machine confidently negotiated both the two-layer zones and the seam intersections, where thickness reached 10 layers. The needle did not break; stitch quality remained high throughout, without skips at thickness transitions.

Test No. 4: Natural Heavy-Weight Leather — 2 Layers

Two layers of thick natural leather were sewn without skips or breaks. This is direct proof that the needle no longer participates in loop tightening — it was precisely this factor that rendered the traditional chain stitch unsuitable for leather.

Test No. 5: Ultra-Light Textile — 2 Layers

The machine sewed the finest material (organza) without gathering or puckering. Tension adjustment was not changed since the previous test on thick leather — a demonstration of true universality.

Test No. 6: Medium-Weight Knitwear — 2 Layers

Knitwear was sewn to high quality; the seam retained the elasticity characteristic of the chain stitch. Seam stretch exceeded 100% elongation — unachievable for the lockstitch.

Test No. 7: Medium-Weight Textile — from 2 to 14 Layers

The machine operated confidently across the full thickness range, confirming that ZARIF 2025 technology is insensitive to package thickness over a wide range. One setting for the entire spectrum.

Test No. 8: Combination — Textile + Leather + Plastic + Plastic Zip

The most complex test: simultaneous sewing of textile, natural leather, plastic film and a plastic zip. Traditional chain stitch would have produced multiple breaks and skips due to differing material friction coefficients. ZARIF 2025 completed the task without a single failure.

Test No. 9: Heavy Denim — 10 Layers: End-of-Seam Tacking with 0.5 mm Stitches

On a package of 10 layers of denim, a tack of 6–10 stitches of 0.5 mm length was performed (total tacking zone length — only 3–5 mm). The seam was so securely fastened that it could not be unravelled by hand even under active mechanical force. The **world's first successful execution of a chain stitch tack with a stitch length of 0.5 mm on such thick material.**

Test No. 10: Denim — 2, 6 and 14 Layers: Beginning-of-Seam Tacking with 0.5 mm Stitches

The beginning of the seam was also tacked with short stitches. No unravelling occurred; the seam held securely, confirming the full suitability of the technology for structurally important seams requiring tacking on both sides.

5.4. Thickness Measurement: Figures That Speak for Themselves

One of the most impressive indicators of ZARIF 2025 is the ability to sew materials up to 8 mm thick with a needle bar stroke of only 32 mm. For comparison:

Technology	Required Needle Bar Stroke for 8 mm Materials	Comparison with ZARIF 2025
Lockstitch (type 301)	36–42 mm	25–33% greater
Traditional chain stitch (type 401)	42–48 mm	30–50% greater
ZARIF 2025	32 mm	Smallest stroke in its class

The short needle bar stroke reduces vibrations, decreases head dimensions and opens the path to high-speed robotisation — a critical advantage for the autonomous factories of the future.

5.5. What Is Not Shown on Video — and Why

The stitch-formation zone beneath the needle plate and the construction of the bottom thread take-up are deliberately not shown in close-up. This is a conscious intellectual property protection strategy, pending completion of patenting of the three key inventions of 2025: the base sewing technology, the automatic thread-trimming device, and the 360° circular sewing mechanism.

Kinematic diagrams of the principal assemblies are available on the website www.zarif.uz in the presentations section, but the precise geometry of components, cam profiles, specific angles and radii will remain trade secrets until patents are obtained.

5.6. The Principal Conclusion from the Tests

✓ DEFINITIVE PROOF:

IF THE ZARIF 2025 PROTOTYPE, IN ITS WORN STATE (play, enlarged clearances, no adjustments), is capable of stably sewing materials from 0.1 mm to 8 mm without a single skipped stitch or thread break, performing reliable tacking with 0.5 mm stitches and demonstrating high seam quality on all types of material, THEN industrial versions with normal tolerances and new components will operate with exceptional stability and longevity.

The wide tolerances of ZARIF 2025 technology mean that the machine maintains high sewing quality even as mechanisms naturally wear. When traditional machines already require an overhaul, ZARIF continues to operate like new. This is not conjecture — it is a fact proved by 28 years of operation of a single prototype.

CHAPTER 6. THE INDUSTRIAL ECOSYSTEM OF THE FUTURE

Dry Head Architecture, Digital Control and Dual-Purpose Machines

◈ ENGINEERING CONCEPT — INDUSTRIAL VERSION (NOT IMPLEMENTED)

This entire chapter describes the concept of the industrial version of ZARIF 2025. The prototype operates on cast-iron plain bearings with manual lubrication. The industrial version with DLC bearings, PEEK bushings and dry operation has not yet been created. The components described are commercially available series products on the market in 2026.

6.1. Why Cast Iron and Oil Are Now a Thing of the Past

The ZARIF 1997 prototype proved the viability of the technology, but its construction was a compromise. The main shaft rotated in cast-iron plain bearings lubricated manually. For laboratory testing this was sufficient, but for industrial operation — categorically not. At 5,000 rpm, cast iron without oil seizes within minutes. Oil mist inevitably contaminates the fabric, which is unacceptable for light-coloured products and medical textiles.

Creating the industrial version of ZARIF 2025 requires a complete rethinking of the approach to materials and lubrication. The goal is a "dry" head (Dry Head), operating at 5,000 stitches/min without a single drop of oil, with minimal wear and maximum cleanliness.

6.2. Key Engineering Solutions for the Industrial Version

6.2.1. DLC Ceramic Hybrid Bearings for the Main Shaft

A solution derived from the aerospace and high-speed machine-tool industries. The steel bearing rings are coated with Diamond-Like Carbon (DLC) with a hardness of up to 2,500 HV and a friction coefficient of 0.05–0.1, allowing them to operate without liquid lubrication. Silicon nitride (Si_3N_4) ceramic balls are 2.5 times lighter than steel ones, reducing centrifugal forces and eliminating the "welding" effect at high rotational speeds. The separator made of PEEK polymer with graphite inclusions acts as a solid lubricant. The service life of such a construction exceeds 10,000 hours without maintenance.

6.2.2. Composite Connecting Rod and PEEK Needle Bar Bushings

The crank-connecting rod mechanism is one of the principal sources of noise and vibration. In the industrial version, the connecting rod body is manufactured from aircraft aluminium 7075-T6 with anodising, the upper head is fitted with a needle roller bearing with ceramic rollers, and the lower head with a self-lubricating Iglidur X (Igus) polymer liner. The connecting rod mass is reduced by 70%, and inertial loads fall radically. The needle bar bushings are made of PEEK-CF30 (polyether ether ketone reinforced with 30% carbon fibre) — a material with stiffness close to aluminium and excellent wear resistance in dry operation.

6.2.3. Thermodynamics of the Chain Stitch: Built-In Cooling

The rotary looper at 10,000 rpm (2:1 transmission ratio) acts as a fan blade. This effect is used for active cooling: the inner cavity of the housing is designed as a sealed circuit, and the air driven by the looper passes through labyrinthine channels, flows around the heating components and exits through micro-apertures, creating positive pressure that prevents dust ingress. A guide nozzle focuses the airflow on the needle shank, cooling it and preventing thread overheating when sewing leather 8 mm thick.

6.3. Digital Control: Full Command over Every Stitch

The industrial version of ZARIF 2025 is designed as a fully digital device. All sewing parameters are controlled programmatically via stepper motors and servo drives:

Parameter	Implementation Technology	Market Status
Stitch length (0.5–5 mm)	NEMA 23 stepper motor on the eccentric adjustment mechanism of the feed dog	Analogues at Juki, Brother, Dürkopp Adler
Top thread tension	Electromagnetic or stepper tensioner with encoder; range 0–500 g, precision ±5 g	Ready market solution
Bottom thread tension	Analogous electronic tensioner on the bottom thread path	Ready market solution
Presser foot pressure	PM35S-048 stepper motor changing spring tension; feedback from foot height sensor	Digital foot mechanism implemented by several manufacturers
Sewing speed (50–5,000 stitches/min)	Built-in servo motor controller with PID regulation; rotor encoder feedback	Standard function of all modern servo motors

Needle positioning	EPS (Electronic Positioning System) — angular encoder on motor shaft; stop in top or bottom position	Standard function

6.4. Sensor Ecosystem: The "Sensory Organs" of a Smart Machine

For autonomous operation and predictive maintenance, the industrial version is equipped with a complex of sensors transmitting telemetry at up to 1 kHz:

- **Piezoelectric thread-break sensors (top and bottom)** — detect cessation of micro-vibrations in the moving thread; ready-made Eltex solutions.
- **Optical knot-detection sensor** — a narrow-focus IR beam monitors thread diameter; stops the machine before the knot reaches the needle eye.
- **Magnetic presser foot height sensor (AS5311, precision 0.01 mm)** — determines material thickness in real time and automatically adjusts presser foot pressure.
- **Servo motor load cell** — current monitoring for measurement of needle penetration force; overload protection, predictive analytics.
- **Thread tension strain gauges** — continuous monitoring to within ±2 grams; feedback for digital tensioning devices.

6.5. The Dual-Purpose Machine Concept

One of the key strategic ideas of the ZARIF 2025 platform: the same machine can work both with a human operator and with a humanoid robot. This allows factories to transition to autonomy in stages, without complete equipment replacement.

👤 Human-Operator Mode	🤖 Humanoid Robot Mode
Control via traditional foot pedal	Pedal excluded from the control circuit
7–10 inch touch display with intuitive interface	All commands via ROS 2 API
Voice control for hands-free operation	Real-time synchronisation with robot (1 ms cycle)
Automatic tacking and thread trimming	Full access to telemetry for robot AI
Laser seam-line indicator	Digital presets for material change in 2–3 seconds
AI assistant suggests optimal settings	Machine becomes a robot "peripheral device"

Principal advantage: capital investments are protected. A factory can begin with regular operators on ZARIF 2025 machines, obtaining an immediate productivity gain, and gradually introduce robots to the same machines. No "instant leap" to autonomy is required.

6.6. The Automatic Thread-Trimming System: The Key Node of Autonomy

Without reliable automatic thread trimming, autonomous production is impossible — every few minutes the robot would be forced to stop for manual trimming. The device, invented in 2020 and refined in 2025, is specifically designed for the rotary looper and has no equivalent anywhere in the world:

- **Trimming beneath the needle plate** — blades in the stitch-formation zone; top and bottom threads trimmed simultaneously.
- **Automatic retention of the bottom thread end** — after trimming the end is held by a leaf spring; re-threading is not required.
- **Mechanical seam-end lock** — the top thread end is passed through the last bottom thread loop, creating a lock that makes unravelling physically impossible.

◈ **STATUS:**

Full-scale 1:1 drawings complete. Prototype not yet manufactured. The principal task of this stage is to bring the device's reliability to 1 million trouble-free operating cycles.

6.7. The 360° Circular Sewing Mechanism: An Alternative to Expensive Automatics

Traditional pattern-sewing automatics with a rotating head are extremely complex, expensive ($25,000–35,000) and limited in speed (2,000–2,500 stitches/min). Their high cost is determined by the need for rigid synchronisation of the entire head's rotation with the material feed direction. ZARIF 2025 technology offers a fundamentally different solution: the machine head and all lower mechanisms (rotary looper, bottom thread pusher, cam bottom thread take-up, automatic trimming device) remain stationary.

Stitch formation in any direction (0–360°) is achieved thanks to a special "circular sewing mechanism" — an innovative unit developed specifically for the ZARIF 2025 platform. Invented in 2020 and refined in 2025, this mechanism is added to the automatic's construction and allows the single-direction stitch-formation technology to operate through the full circumference of the needle hole in the needle plate. The material moves on a CNC X-Y table, whilst stitch quality remains consistently high at any feed angle.

◈ **STATUS:**

Full-scale 1:1 drawings theoretically demonstrate the possibility of stitch formation in any direction. The mechanism prototype has not yet been manufactured. It remains necessary to manufacture a prototype automatic, confirm the mechanism's viability in practice, and achieve the required reliability through multi-cycle testing on various materials.

Parameter	Traditional Automatic with Rotating Head	ZARIF Automatic with Circular Sewing Mechanism
Maximum speed	2,000–2,500 stitches/min	Up to 3,500 stitches/min
Head mass	40–60 kg	15–20 kg (reduction of 60–70%)
Noise level	75–85 dB	60–65 dB
Annual maintenance costs	$2,000–3,100	$500–800
Payback period	8–12 years	3–5 years
Equipment cost	$25,000–35,000	$15,000–20,000 (expected)

CHAPTER 7. INTEGRATION WITH HUMANOID ROBOTS AND ARTIFICIAL INTELLIGENCE

ROS 2, OPC UA, NVIDIA Jetson Thor and Three Levels of Robotisation

◈ **ENGINEERING CONCEPT — INDUSTRIAL VERSION (NOT IMPLEMENTED)**

The ZARIF 2025 prototype has no digital interfaces or stepper motors. All protocols, APIs, synchronisation algorithms and integration methods described are an engineering concept of the industrial version.

✓ MECHANICAL READINESS FOR ROBOTISATION IS PROVEN

The key barriers for robots — the bobbin, small clearances, unpredictable skipped stitches and thread breaks — have been eliminated in the ZARIF 2025 mechanics. This is confirmed by the prototype. It is precisely this that makes digital integration realistic for the first time.

7.1. Why Traditional Machines Destroy Robotisation

A humanoid robot costing $30,000–$250,000, placed at a traditional lockstitch machine, becomes a bobbin-replacement operator. Every 12–20 minutes it is forced to stop work, remove the bobbin case, change the bobbin, re-thread with sub-millimetre precision. Manual tension adjustment, replacement of a broken needle, re-threading after a break — all of this requires human dexterity and tactile feedback that robots do not yet possess. As a result, the robot's effective productivity falls to 45–50%.

Traditional chain stitch dispenses with the bobbin but introduces its own problems: a loose seam, impossibility of reliable tacking, risk of unravelling, frequent skips on dense materials. Neither traditional technology provides the predictability necessary for autonomous 24/7 operation.

7.2. ZARIF 2025 as a Robot-Native Platform: Mechanics Resolve Everything

Before speaking of digital protocols, it is important to understand: the principal obstacle to robotisation was always mechanical. ZARIF 2025 eliminates it at the hardware level:

- **No bobbin** — the bottom thread is supplied from a 1–5 kg cone. The robot can operate for 8–24 hours without intervention.
- **No critical clearances** — needle-looper clearance 0.5 mm (5–10 times greater than traditional). Needle deflection does not lead to collision.
- **Guaranteed stitch formation** — skips and thread breaks are practically absent. The robot's AI need not constantly "await a failure" and be ready to intervene.
- **No adjustments on material change** — one digital preset for any material from the 0.1–8 mm range.

7.3. Digital Interaction Architecture

Level	Protocol	Frequency	Purpose
High-level commands	ROS 2 (Topics/Services)	Event-driven	Start/stop, programme loading, digital preset change
Motion synchronisation	EtherCAT	1 kHz (1 ms)	Real-time robot–machine coordination
State exchange	OPC UA / MQTT	100 Hz (10 ms)	Telemetry, sensor status, MES/ERP integration
Machine vision	Shared Memory / DDS	30 Hz (33 ms)	Video streams and analysis results from NVIDIA Jetson

7.4. Decision-Making Hierarchy: Who Is in Charge?

In the "humanoid robot — sewing machine" pair, a Master–Slave architecture is established, with the machine acting as the master device:

- **Machine AI (Master)** — manages the stitch-formation process, monitors quality in real time, adapts parameters (speed, tension, presser foot pressure) based on sensor data.

- **Robot AI (Planner)** — determines the seam trajectory, manages material feed, coordinates grips and component rotations.
- **Joint decision-making** — when an anomaly is detected (e.g., approach of a thickened seam), the machine AI and robot AI jointly change parameters: the robot adjusts feed speed, the machine adjusts stitch length and presser foot pressure.

7.5. AI-to-AI Command Language

For direct machine control by the robot, a protocol based on JSON commands transmitted via ROS 2 has been developed. Below are examples of key commands.

START_SEWING — begin sewing with automatic tacking:


```
{
  "command_type": "START_SEWING",
  "parameters": {
    "backtack": { "enabled": true, "stitch_count": 8, "stitch_length": 0.5, "speed": 200 },
    "initial_speed": 200, "target_speed": 3000,
    "acceleration_profile": "SMOOTH_S_CURVE", "acceleration_time": 1.5,
    "thread_tension": { "upper": "AUTO", "lower": "AUTO" },
    "presser_foot_pressure": "AUTO", "needle_stop_position": "DOWN"
  }
}
```


The machine performs 8 tacking stitches of 0.5 mm length, then accelerates smoothly to 3,000 stitches/min, holding the needle in the down position during pauses.

PAUSE_SEWING — pause for material rotation:


```
{
  "command_type": "PAUSE_SEWING",
  "parameters": {
    "needle_position": "DOWN_IN_FABRIC",
    "presser_foot_action": "REDUCE_PRESSURE", "pressure_reduction": 70,
    "maintain_thread_tension": true, "timeout": 30.0
  }
}
```


The machine stops with the needle inside the material, reduces presser foot pressure by 70% (so the robot can easily rotate the component), maintaining thread tension to prevent seam unravelling. After rotation the robot sends RESUME_SEWING.

STOP_SEWING — completion with tacking and automatic thread trimming:


```
{
  "command_type": "STOP_SEWING",
  "parameters": {
    "backtack": { "enabled": true, "stitch_count": 10, "stitch_length": 0.5, "speed": 200
},
    "thread_cutting": { "enabled": true, "cut_after_stitch": 11, "lower_thread_retention":
"SPRING_CLAMP" },
    "presser_foot_action": "LIFT_FULL", "needle_position": "UP"
  }
}
```


10 tacking stitches of 0.5 mm, then automatic thread trimming under the plate, retention of the bottom thread end by spring clamp, presser foot lift, needle stop in the up position. The seam is secured by a mechanical lock — unravelling is impossible.

7.6. Real-Time Synchronisation and the Anti-Deflection Algorithm

Continuous telemetry at 1 kHz (via EtherCAT) allows the robot to "feel" the sewing process. The machine transmits: current sewing speed, needle position in the cycle, both thread tensions, material thickness under the foot, needle penetration force, and AI stitch quality analysis results.

Based on this data, the **Anti-Deflection algorithm** is implemented — intelligent prevention of needle bending and breakage when crossing thickened seams:

Time	System Action
T = 0 ms	Robot and machine cameras detect a thickened seam ahead on the trajectory.
T = 33 ms	AI decides to reduce stitch length so the needle penetrates the material 1 mm before the seam edge.
T = 70 ms	Machine adjusts stitch length, reduces presser foot pressure and increases servo motor torque to ease the climb onto the thickening.
T = 75 ms	After crossing the thickening, previous stitch length and presser foot pressure are restored. The feed dog temporarily deflects to ease material feed.
T = 125 ms	The needle makes a compensatory "jump" across the edge of the descent — the system ensures penetration occurs 1 mm after the edge.

The entire cycle takes less than 1 second. The needle never penetrates the critical edge of the height transition, completely eliminating deflection, plate collision and breakage.

7.7. Three Types of Integration: A Staged Path to Autonomous Sewing Production

The transition to fully autonomous sewing production requires a clear strategy for the interaction of humanoid robots with future ZARIF 2025 digital machines and automatics. The ZARIF 2025 platform defines three types of such integration — not as isolated variants, but as sequential stages of increasing autonomy.

7.7.1. Type I: Humanoid Robot and ZARIF 2025 Industrial Sewing Machine

The most complex, but also the most versatile type of integration. The robot directly participates in the sewing process: picks up the component, feeds it to the working zone, guides it along the seam line, turns it at corners and curved sections. The machine forms the stitch, whilst the robot performs all the manipulations normally performed by a human operator.

The key to success here is deep digital coordination. Future ZARIF 2025 machines are designed so that the robot receives not only start and stop commands but full access to sensors, machine vision and intelligent functions. The machine can independently, on command from the AI or the robot itself, change stitch length, presser foot pressure, sewing speed, needle position and other parameters in real time.

Particular attention is devoted to preventing needle deflection. The two principal causes of deflection are: penetration of the material at the edge of a thickening when climbing onto it, and when descending from it. In the digital ZARIF machine this problem is resolved algorithmically: on command from the robot, the stitch length is adjusted so the needle never lands precisely at the critical edge of a height transition. When crossing extremely dense packages, the main shaft servo motor automatically increases torque, ensuring stable penetration without increasing the needle bar stroke.

For sewing on tight curves, the machine can automatically reduce presser foot pressure by 70–80% during a pause with the needle in the material, allowing the robot to rotate the component freely. Stitch length on such sections can be temporarily reduced to 1.5–2 mm to maintain seam quality.

Similar digital algorithms support the formation of angular seams without deterioration of the reverse side — another task not soluble by traditional means. When sewing an angular seam using chain stitches, reverse-side seam quality can deteriorate because during material rotation around the needle, threads may become partially trapped beneath the presser foot and material. In future ZARIF machines this can be resolved by a special digital algorithm: the last stitch before the corner point is reduced to 1 mm; the machine stops with the needle inside the material; presser foot pressure is reduced to zero; the humanoid robot rotates the material through the required angle; pressure is restored; the first stitch after the turn is made 1 mm in length; and only then does sewing continue at the normal stitch length.

This type of integration is addressed to the most flexible production facilities with frequent assortment changes, many complex curved seams and a high proportion of unique operations. It is here that the dual-purpose machine concept is fully realised: the same ZARIF machine can work with a human today and with a robot tomorrow, without equipment replacement.

7.7.2. Type II: Humanoid Robot and ZARIF 2025 Pattern-Sewing Semi-Automatic

In the second type of integration, the robot does not participate directly in the sewing process. Its role shifts towards preparation and template servicing: separating the component from the stack, placing it in the template, clamping it, installing the template on the automatic's carriage, starting the cycle, and upon completion — removing the template, releasing the finished product and laying it for further processing.

The sewing operation itself is performed fully automatically by the ZARIF 2025 pattern-sewing automatic. Thanks to the circular sewing mechanism forming the stitch in any direction without rotation of the head and lower mechanisms, the trajectory is processed with high repeatability regardless of contour complexity.

The industrial value of such a cell is determined not only by sewing speed but also by template change speed. Future ZARIF semi-automatics are therefore designed in conjunction with intelligent tooling: rapid clamping, magnetic or combined template-fixing devices, correct-installation sensors and automatic or semi-automatic template-ejection mechanisms.

This type is targeted at operations where the component is readily templateable, and requirements for precision and repeatability are high. It makes significantly lower demands on robot dexterity and can be implemented at earlier stages of automation.

7.7.3. Type III: Humanoid Robot and ZARIF 2025 Automated Sewing System

The third type represents the simplest variant of interaction for the robot. The ZARIF 2025 automated sewing system has a dedicated loading zone. The robot places a component or set of components in it, after which the system independently grips them, moves them to the working zone, performs the sewing and delivers the finished product to an unloading zone. The robot need only collect the finished product and pass it to the next operation or to the in-shop transport.

Here the robot acts not as a "tailor" but as an automated line operator, performing loading, unloading and servicing functions. Requirements for its sensing, coordination and response time are minimal, and cell productivity is most strongly determined by the capabilities of the system itself.

This type is most effective in conditions of large-batch, geometrically stable production. It creates a practical basis for pilotless flow sections where the human's role is limited to monitoring and servicing several cells simultaneously.

7.7.4. Unified Strategy: Three Types as Stages of Development

The three-level model proposed is not merely a classification — it is a ladder by which production can ascend gradually, commensurate with its tasks and economic capabilities. A factory can begin by automating the simplest operations (Type III), work through robot interaction with pattern automatics (Type II), and only then proceed to the most complex but most flexible interaction — robot with industrial machine (Type I).

Each type of integration rests on the same technological core — the ideal ZARIF 2025 sewing technology, fully digital control, and open ROS 2 and OPC UA interfaces. Equipment investment thus remains protected, and the platform itself becomes the foundation upon which autonomous sewing production of any scale can grow.

◈ CONCEPT:

All three types of integration described are an engineering concept of the industrial version of ZARIF 2025. They will be implemented after creation of the first industrial prototype and confirmation of the "machine — robot" digital interaction interfaces.

7.8. Recommended Robotic Platforms

Model	Manufacturer	Key Features	Approximate Cost
Optimus Gen 3	Tesla	22 degrees of freedom per hand, tactile sensors, mass production	$20,000–30,000
Figure 03	Figure AI	Adaptive fingers, 16 degrees of freedom, Helix AI for real-time learning	$30,000–50,000
Unitree G1	Unitree	23–43 degrees of freedom, affordable price	~$16,000
Walker S2	UBTECH	Deep integration in automotive assembly, high autonomy	$20,000–80,000

◈ CONCEPT:

All digital interfaces, protocols and algorithms described are an engineering concept of the industrial version of ZARIF 2025. They will be implemented after attracting Round A funding and creating the first industrial prototype.

CHAPTER 8. THE ZARIF AUTONOMOUS FACTORY

Robotic Trolleys, Lights-Out Production and the Digital Twin

◈ ENGINEERING CONCEPT — INDUSTRIAL VERSION (NOT IMPLEMENTED)

The ZARIF Autonomous Factory has not been built. The robotic trolleys, lights-out architecture and digital twin described are an engineering concept based on commercially available components (AGV platforms, NVIDIA Jetson, LiDAR, UWB localisation) and already implemented principles from adjacent industries (additive manufacturing).

8.1. From Conveyor to Swarm: Why the Autonomous Factory Needs Robotic Trolleys

When unmanned production is discussed, one typically imagines robots at sewing machines. But the most challenging question is logistics. How do cut components travel from the cutting room to the correct sewing cell precisely on time? How do finished products move to packing? How does the system know where an empty trolley is needed and where a loaded one is required?

The traditional answer is a centralised conveyor system: an overhead monorail (UPS) or belt conveyor. Such systems are expensive (installation for a mid-size shop floor: $200,000–$1,000,000), completely inflexible (changing a route requires physical rebuilding) and create a hard dependency of the entire production on a single link — if the conveyor breaks down, the entire factory halts.

The ZARIF Autonomous Factory takes a fundamentally different path: **decentralised logistics based on autonomous mobile robots (AMR).** Each robotic trolley is a self-contained intelligent agent that independently plots its own route, communicates with robots and machines, and carries production assignments in its own memory. No central conveyor, no single point of failure.

8.2. The ZARIF Robotic Trolley: Intelligence on Wheels

The ZARIF Robotic Trolley is not merely a transport platform. It is a fully autonomous mobile robot, inspired by the Tesla Autopilot architecture. Each trolley is equipped with:

- A mini-supercomputer (NVIDIA Jetson Orin NX or equivalent) into whose memory the factory's production projects are loaded. The trolley "knows" which components to deliver where, in what sequence, and with which robot to interact.
- Stereo cameras and LiDAR for SLAM navigation — simultaneous mapping of the shop floor and localisation. The trolley builds a real-time map, updates it when equipment is repositioned, and plots optimal routes avoiding obstacles.
- An AI identification module that recognises humanoid robots, sewing machines and other trolleys via visual markers, RFID tags or through direct communication via ROS 2.
- Wireless recharging — the trolley automatically proceeds to a charging station when battery level falls, without requiring human intervention.

The factory's central server performs only monitoring and strategic planning functions. Each trolley makes decisions independently, providing fault tolerance unachievable by centralised systems.

8.3. The Dual-Binding Principle: Zero Seconds of Downtime

The key algorithmic innovation of the ZARIF Robotic Trolley is the principle of dual (and sometimes triple or quadruple) binding to technological cells. This algorithm guarantees that neither robots nor trolleys ever wait idle for each other.

Consider a simple example: Cell No. 3 is attaching pockets; Cell No. 4 is joining side seams. Semi-finished products must flow from Cell 3 to Cell 4.

- Trolley A stands at Cell 3, loaded and ready to depart.
- Trolley B stands at Cell 4, empty and ready to receive.
- As soon as the robot at Cell 4 takes the last semi-finished product from Trolley B, it independently travels to Cell 3.
- Trolley A, seeing that the space at Cell 4 has been freed, makes its way there.
- The freed Trolley B approaches Cell 3 and takes the place of Trolley A.

The cycle repeats continuously. Trolleys are in constant motion, and robots never wait for components. OEE reaches 95%+ — a figure unthinkable for conveyor systems.

8.4. Interchangeable Modules: Universality in Seconds

Each ZARIF Robotic Trolley consists of a universal mobile platform (lower block) and an interchangeable upper module. Module change is performed by maintenance personnel in seconds outside the production cycle. The lower block automatically determines the type of installed module via RFID tag and adapts its movement algorithms accordingly (speed, braking distance, priority).

Module	Description	Application
Cut tray	Flat tray with retainers, components stacked in even layers	Cutting room → sewing room
Component stack	Vertical dividers for bundles of components of different sizes	For cut components
Hanger (GOH)	Horizontal GOH-type rails	Finished products → warehouse
Delicate	Soft cradles with thermal protection	Silk, knitwear, hot components after steam-pressing
Roll (up to 250 kg)	Horizontal axle with retaining lock	Fabric rolls: warehouse → cutting room
Knit bale	Wide flat platform with side guards	Fabric bales → spreading machine
Trimmings	Cell organiser with RFID tags	Buttons, zips, labels, thread

8.5. Comparison with Traditional Systems

Criterion	Overhead Rail System (UPS) / Conveyor	ZARIF Robotic Trolley
Installation and infrastructure	Design, welding, power supply throughout the route. Cost $200,000–$1M+. Months of work.	No fixed infrastructure required. Floor marking + software setup. Days, not months.
Route flexibility	Route fixed by the rail. Change = physical rebuilding.	Route reprogrammed automatically when production project changes. Absolute flexibility.
Control intelligence	Centralised controller. Failure = entire production stops.	Decentralised AI on each trolley. Failure of one does not affect the others.
Robot interaction	Passive: delivers component and stands idle.	Active: trolley identifies its robot and communicates via AI protocol.
Scaling	Adding machines = extending the monorail.	Adding machines = mark a space on the floor + update the digital map.

8.6. Architecture of the ZARIF Autonomous Factory: Three Levels of Integration

The ZARIF Autonomous Factory is a cyberphysical system where physical processes (cutting, sewing, steam-pressing, packing) and digital processes (MES, ERP, digital twin, AI control) merge into a unified whole. The architecture is built on three levels:

Level 1: Periphery (Edge). Every device — ZARIF 2025 sewing machine, humanoid robot, robotic trolley, cutting complex — has its own embedded computer and operates autonomously, exchanging data in real time. Sewing machines transmit telemetry (stitch count, thread tension, needle load, quality of each stitch via machine vision). Robots transmit information on component gripping and cycle time. Trolleys report location, load and battery status.

Level 2: Manufacturing Execution System (MES). MES collects data from all devices, tracks order fulfilment in real time and manages material flows. Each cut component receives an RFID tag associated with the model, size, colour and process specification. The trolley collecting the component automatically "knows" which cell to deliver it to. When a cell fails, MES redistributes the workload to reserves and trolleys automatically change their routes. AI cameras above each machine analyse every stitch; on detection of a defect, the system stops the machine and flags the product.

Level 3: ERP and Business Logic. The ERP system receives customer orders, manages raw material stocks, plans capacity loading and passes production assignments to MES. The factory's digital twin allows simulation of a new product launch, identification of bottlenecks and layout optimisation before physical implementation.

8.7. Lights-Out Production: 24/7 Without Personnel on the Shift Floor

The ZARIF Autonomous Factory is designed for "lights-out" operation — without personnel on the production floor. This is achieved through five factors that together create an autonomous environment:

1. **No bobbin.** The principal barrier to continuous operation is eliminated. Thread cones are changed by robots or technicians once every 8–24 hours.
2. **Automatic thread trimming and threading (ZARIF 2025 concept).** The auto-trimming device cuts threads and holds the bottom thread end by spring. The start of the next cycle requires no manual threading.
3. **Self-diagnostics and predictive maintenance.** AI analyses vibrations, bearing temperatures and servo motor currents — predicts when needle replacement or maintenance will be needed. A technician is called only when necessary.
4. **Robotised logistics.** All material movements are performed by ZARIF Robotic Trolleys: they self-charge, select routes and synchronise with humanoid robots.

5. **Automatic programme changes.** When switching to a new order, MES transmits new parameters to all devices in seconds. No physical retooling required.

8.8. The EP-M300L Precedent from Eplus3D: Proof of Feasibility

One of the most frequent questions is: "This sounds impressive, but is it realisable in practice?" The answer is yes, and it has already been realised in an adjacent industry. On 5 March 2026, Eplus3D published a description of its EP-M300L Production-Ready Automation Line — a turnkey solution for high-volume metal 3D printing already deployed for clients in more than 50 countries.

EP-M300L uses removable build cylinders replaced as a single unit without downtime; links key processes via AGV without human involvement; is operated by a single operator managing several lines; integrates with MES and generates a "digital birth certificate" for every component.

✓ PRECEDENT:

EP-M300L proves that autonomous production with AGV logistics, decentralised control, closed-loop MES quality monitoring and lights-out mode is not futurology but operational reality. ZARIF creates for sewing production what EP-M300L created for metal 3D printing: a deterministic, reliable, robotisable working tool.

8.9. The Digital Twin in NVIDIA Omniverse: Virtual Training Accelerates Launch

The ZARIF Autonomous Factory uses the NVIDIA Omniverse / Isaac Sim platform to create a complete digital twin that performs critical functions:

- **Production project simulation:** before launching a new product, engineers load CAD models of components, the process specification and sewing parameters. The system simulates the entire flow and identifies bottlenecks. Optimisation takes hours, not weeks of real production downtime.
- **Humanoid robot training (sim-to-real):** training a robot to grip fabric in the real world is slow and risky. In the digital twin, the robot completes thousands of cycles, accumulating experience that is then transferred to the physical robot. Commissioning time is reduced from months to days.
- **Predictive AMR fleet calculation:** for each project the digital twin computes the optimum number of robotic trolleys, their distribution across cells and their recharging schedule.
- **Real-time OEE monitoring:** once the physical factory is launched, the digital twin runs in parallel, receiving data from all devices and comparing actual output with the plan.

◈ CONCEPT:

All components of the ZARIF Autonomous Factory — robotic trolleys, MES integration, lights-out architecture, digital twin — exist as a thoroughly developed engineering concept, based on technologies available on the 2026 market and a confirmed precedent from an adjacent industry. Implementation will begin after creation of the industrial version of the ZARIF 2025 machine.

CHAPTER 9. ECONOMICS AND PAYBACK

How ZARIF 2025 Reduces OPEX by 75%, Increases OEE to 85%+ and Achieves ROI in 11.1 Months

◈ ENGINEERING CONCEPT — INDUSTRIAL VERSION (NOT IMPLEMENTED)

All economic calculations in this chapter are based on the parameters of the conceptual ZARIF Autonomous Factory and projected component prices for 2026–2028. Actual figures will be determined after creation of the industrial version and production testing. The calculations demonstrate the potential of the technology upon its full realisation.

9.1. Input Data: A Typical Sewing Enterprise

For comparative analysis, a typical medium-sized sewing enterprise is taken: 100 workstations, two-shift operation (16 hours per day, 250 days per year), production of mid-price-segment clothing. Figures are based on real industry data for 2024–2026, consultations with production specialists and analytical reports.

Parameter	Traditional Factory	ZARIF Autonomous Factory
Machine type	Lockstitch (301) + traditional chain stitch (401)	ZARIF 2025 (single-needle, pattern, specialised)
Operating mode	2 shifts × 8 hrs = 16 hrs/day, 250 days/yr	24 hrs/day, 365 days/yr
Personnel (operators)	100 persons	0 (full autonomy)
Personnel (technicians)	15 persons	3 persons (remote monitoring)
Average machine cost	$2,500 (lockstitch) + $3,500 (chain stitch)	$8,000 (ZARIF 2025, weighted average)
Average machine power	400 W	300 W
Defect rate	3–5% (industry average)	<0.1% (AI control of every stitch)
Annual output (items)	500,000	985,000

9.2. CAPEX Calculation (Capital Expenditure)

Item	Traditional Factory	ZARIF Autonomous Factory
Machines	100 × $2,500 = $250,000	100 × $8,000 = $800,000
Infrastructure (tables, lighting, ventilation)	$150,000	$150,000
AGV / logistics	$0 (manual trolleys)	$800,000 (40 trolleys × $20,000)
Humanoid robots	$0	$1,500,000 (30 robots × $50,000)
AI / MES / camera systems	$0	$300,000
Personnel training	$50,000	$50,000
Contingency fund	$50,000	$200,000
TOTAL CAPEX	**$500,000**	**$3,800,000**

The CAPEX of the autonomous factory is 7.6 times higher. This is the principal barrier to decision-making, which is overcome by analysing operating costs and productivity.

9.3. OPEX Calculation (Annual Operating Costs)

Item	Traditional Factory	ZARIF Autonomous Factory
Operator wages	100 persons × 16 hrs/day × 250 days × $15/hr = $6,000,000	$0

Technician wages	15 persons × 8 hrs/day × 250 days × $25/hr = $750,000	3 persons × 24 hrs/day × 365 days × $30/hr = $788,400
Social contributions (30%)	$2,025,000	$236,520
Electricity	100 machines × 0.4 kW × 4,000 hrs × $0.12 = $19,200	100 machines × 0.3 kW × 8,760 hrs × $0.12 = $31,536
Thread	500,000 items × 100 m × $5/kg / 10,000 = $25,000	985,000 items × 95 m × $5/kg / 10,000 = $46,787
Spare parts and servicing	$30,000	$80,000
Defective product write-offs	4% × 500,000 × $5 = $100,000	0.1% × 985,000 × $5 = $4,925
Rent / depreciation	$150,000	$150,000
Miscellaneous expenses	$100,000	$150,000
TOTAL OPEX	**$9,199,200**	**$1,488,168**

ANNUAL OPEX SAVING: $9,199,200 − $1,488,168 = $7,711,032

9.4. Productivity and Revenue

Indicator	Traditional Factory	ZARIF Autonomous Factory
Output (items/year)	500,000	985,000 (+97%)
Average sale price (wholesale)	$50	$50
Annual revenue	$25,000,000	$49,250,000
Revenue differential	—	+$24,250,000/year

9.5. Payback and NPV

Additional investment (CAPEX differential): $3,800,000 − $500,000 = $3,300,000.
Annual OPEX saving: $7,711,032.
Additional revenue: $24,250,000.
Total additional EBITDA: $31,961,032. After tax (20%): $25,568,826.

CONSERVATIVE CALCULATION (OPEX saving only, excluding revenue growth):
$3,800,000 / ($4,120,000 / 12) ≈ **11.1 months — payback period on additional investment.**

NPV over 10 years (discount rate 10%):

- Accounting for OPEX saving only: ≈ $16.4 million
- Accounting for additional revenue: $34.6 million and above

9.6. 197% Productivity: How It Is Achieved

The output growth of 97% (to 197% of base level) is not magic — it is the direct result of four factors:

1. **24/7/365 operation.** The autonomous factory operates 8,760 hours per year against 4,000 hours for a two-shift operation. Even allowing 5% time for maintenance, this yields almost twice the useful operating time.

2. **Reduction in defects from 3–5% to <0.1%.** AI control of every stitch, automatic parameter adjustment and elimination of the human factor. At an annual output of 985,000 items, savings on defective product exceed $200,000.
3. **No downtime for bobbin changing.** In traditional machines this accounts for 3–5% of working time. In ZARIF 2025 — zero.
4. **No adjustments on material change.** Retooling that takes 15 minutes to an hour on a traditional factory is performed in ZARIF in 2–3 seconds by loading a digital profile.

9.7. Comparison of ZARIF 2025 with Competing Automation Solutions

Indicator	SoftWear Automation (Sewbot)	Traditional machines + robot	ZARIF 2025 + humanoid robot
Cost of automated cell	$100,000–350,000	$30,000 (robot) + $6,000 (machine) + bobbin losses	$28,000–36,000
Effective robot productivity	High, but for simple products only	45–50% (due to bobbin stoppages and breaks)	95%+
Product versatility	Simple single-layer items only (T-shirts, pillowcases)	Limited by human or robot capability	Full range: from silk to 8 mm leather
Gradual implementation	Not possible (complete line replacement)	Possible	Built into the architecture ("human" → "robot" mode)
Payback period	5–8 years (limited flexibility)	More than 3 years (low efficiency)	11.1 months

9.8. Summary Economic Results

THE ZARIF AUTONOMOUS FACTORY CONCEPT DELIVERS:

- → OPEX reduction by a factor of 6: from $9.2M to $1.5M per year
- → Output growth of 97%: from 500,000 to 985,000 items/year
- → Defect reduction from 3–5% to <0.1%
- → Payback on additional investment: 11.1 months
- → NPV over 10 years (conservative, OPEX saving only): $16.4M
- → NPV over 10 years (including revenue growth): $34.6M

No other automation project in the sewing industry demonstrates comparable indicators.

◈ NOTE:

Actual figures will depend on the real cost of the industrial version of the ZARIF 2025 machine, the chosen humanoid robot model and specific production conditions. The figures given are a conservative estimate of the technology's potential upon full realisation of the concept.

CHAPTER 10. ROADMAP 2026–2035

From the Flatbed Platform to a Global Autonomous Ecosystem

◈ ENGINEERING CONCEPT — INDUSTRIAL VERSION (NOT IMPLEMENTED)

The roadmap describes the sequence of creating products that do not yet exist. This is a development and commercialisation plan. Each stage creates a standalone commercial product enabling revenue generation to fund the next stage.

10.1. Why the Staged Approach Is the Only Correct One

ZARIF 2025 technology has been maturing for 31 years. The next stage — industrial commercialisation — demands an equally considered approach. It is not possible to build an autonomous factory all at once. Nor is it possible simultaneously to launch the entire product range. But it is possible — and necessary — to create a sequence of products, each with independent market value, each generating revenue and each funding the development of the next.

10.2. Roadmap Overview

Stage	Horizon	Product	Key Task	Market Result
I	2026–2028	Single-needle flatbed machine	Auto-trimming device (1M cycles). Integration of sensors, AI, ROS 2, OPC UA.	World's first dual-purpose machine. Foundation for entire range.
II	2028–2030	Pattern-sewing automatic with 360° mechanism	360° mechanism prototype, multi-cycle testing, CAD-file software.	World's first chain stitch type 401 pattern automatic.
III	2030–2031	Specialised automatics: buttonholes, bar tacks, buttons	Platform adaptation, specialised tooling, unified software.	Full range on a single technological base.
IV	2031–2032	Automated systems for any seams	Machine vision, MES/ERP integration, automatic loading/unloading.	Ready-made modules for autonomous lines.
V	2032–2035	Full range + autonomous factory model roll-out	Modular architecture. ZARIF Autonomous Factory roll-out. OEM licensing.	Industry standard. 15–20% of market ($4–5 Bn).

Stage I (2026–2028): Single-Needle Flatbed Machine

Industrial version of the ZARIF 2025 single-needle sewing machine with flatbed platform and bottom material feed. This is the most in-demand configuration — up to 70% of all industrial machines in the world have precisely this architecture.

Principal technical task: design, manufacture and qualification to a service life of no fewer than 1,000,000 cycles of the automatic thread-trimming device.

Parallel tasks: integration of all sensors into a unified control system; digital regulation of all sewing parameters via stepper motors; installation of mini-supercomputer (NVIDIA Jetson Orin NX) and machine vision cameras; implementation of ROS 2 and OPC UA software interfaces; creation of HMI user interface with touch display and voice control.

Target price: $5,000–6,000 for a fully equipped machine with AI, machine vision and digital control.

Result: the world's first industrial dual-purpose sewing machine — capable of operating both with a human operator and with a humanoid robot (ROS 2). Reliability benchmark: 0 skipped stitches, 0 thread breaks, 0 needle breakages, 0 oil stains, 0 adjustments on material change from silk to 8 mm.

Stage II (2028–2030): Pattern-Sewing Automatic with 360° Mechanism

A ZARIF 2025 pattern-sewing automatic equipped with the innovative circular sewing mechanism. Unlike traditional automatics with a rotating head, in the ZARIF 2025 automatic neither the machine head nor the lower mechanisms rotate. Instead, the special circular sewing mechanism allows formation of quality stitches when feeding material in any direction (0–360°). The material moves on an XY table following a programme read from a CAD file.

Target price: $10,000–15,000 — comparable with standard lockstitch automatics but with fundamentally superior characteristics.

Stage III (2030–2031): Specialised Automatics

Based on the pattern automatic (Stage II), narrow-specialist machines are developed: buttonhole automatics for straight, eyelet and shaped buttonholes; bar-tacking automatics with programmable geometry (rectangular, round, arrowhead); and button-sewing machines for attaching 2-hole and 4-hole buttons and shank buttons. **Key feature:** a unified software platform (ROS 2, OPC UA) for all types of automatics.

Stage IV (2031–2032): Automated Systems

Automated sewing systems for long and short seams on high-volume products: side seam joining of trousers, cuff attachment, shoe upper component joining. The systems include automatic material feed, machine vision for trajectory control and correction, and full MES/ERP integration. These systems serve as a bridge between manual labour and full automation.

Stage V (2032–2035): Full Range and Roll-Out

Range expansion to all platform types: flatbed (all feed types), post-bed (footwear, bags, gloves), cylinder-bed (sleeves, cuffs, trouser legs), roller feed for heavy and elastic materials. Simultaneously — roll-out of the ZARIF Autonomous Factory model: standard projects for various product types, licensing of technology to manufacturers and system integrators.

10.3. Projected Results by 2035

UPON FULL REALISATION OF THE CONCEPT BY 2035:

- ZARIF 2025 — the industry standard for oil-free, bobbin-free industrial sewing.
- The world's first pattern automatic with chain stitch and circular sewing.
- The world's first buttonhole, bar-tacking and button-sewing automatics with the new double thread chain stitch type 401.
- ZARIF Autonomous Factory — a replicable model licensed worldwide.
- 15–20% of the sewing equipment market ($4–5 billion by 2035).
- Marketplace for digital sewing programmes (subscription model).
- A 20-year patent barrier providing protection against copying.

10.4. Stage Funding: The Self-Financing Model

Each stage of the roadmap is designed to generate sufficient revenue to fund the next. Stage I (flatbed platform) enters the market and begins generating income that is reinvested in developing Stage II (pattern automatic). By the time Stage III is launched, the company already has two products on the market and a stable cash flow. This model minimises the need for external funding after the initial Round A and reduces investor risk.

10.5. Key Risks and Mitigation Strategy

Risk	Probability	Mitigation Strategy

Delay in qualifying auto-trimming device (1M cycle life)	Medium	Parallel testing on multiple rigs; time buffer in Stage I schedule (24 months instead of 18)
Patenting difficulties in certain jurisdictions	Low (priority confirmed by WIPO in 1996)	Engagement of local patent attorneys; priority filing in USA, China, EU, Japan, South Korea
Delay in humanoid robot market entry	Low (Tesla, Figure AI already in production)	Machine originally works with human operator; robotisation is an option, not a prerequisite
Emergence of competing technology	Extremely low (28-year head start + patents)	Accelerated patenting of three new 2025 inventions; trade secret regime for know-how

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CHAPTER 11. INVESTMENT OPPORTUNITY

$400–900 Billion Market, 20-Year Patent Protection and Two Proposals for Partners

✓ ALREADY PROVEN BY THE ZARIF 2025 PROTOTYPE

The ZARIF 2025 prototype exists and has practically proved the stitch technology. US Patent No. 6,095,069 is in force. Three new patent applications are being prepared for filing via WIPO (PCT) in 20+ jurisdictions.

◈ ENGINEERING CONCEPT — INDUSTRIAL VERSION (NOT IMPLEMENTED)

The industrial version of the machine, the automatic thread-trimming device and the circular sewing mechanism require creation and testing. It is precisely for this purpose that the investments described in this chapter are intended.

11.1. Market Opportunity: $400–900 Billion over 15 Years

Market	Current Volume	Forecast to 2035	Link to ZARIF 2025
Industrial sewing equipment	$3+ Bn/yr	$4–5 Bn/yr	Direct sales market for ZARIF 2025 machines (replacing 90% of lockstitch operations)
Humanoid robots	Nascent	$38 Bn	Robots receive a native sewing tool — a killer application for manufacturing robotics
AI manufacturing services	$10+ Bn/yr	$50+ Bn/yr	ZARIF — the only native sewing platform for digital twins, predictive analytics and cloud services
World textile market	$2.36 Tn/yr	$3+ Tn/yr	The end market for autonomous factories

TOTAL ADDRESSABLE MARKET (TAM) — 15-YEAR HORIZON:

- Hardware (humanoid robots + ZARIF machines): $250–600 billion
- Software and AI services: $100–200 billion
- Integration and services: $50–100 billion

TOTAL: $400–900 BILLION

11.2. Intellectual Property Strategy: 20-Year Protection

The competitive advantage of ZARIF 2025 rests on three levels of intellectual property protection: patents, trade secrets and know-how.

Patent Portfolio

The base patent US No. 6,095,069 protects the principle of dual-rotation kinematics of the rotary looper. In 2025, three new patent applications were prepared, to be filed via PCT immediately after funding is secured:

1. **ZARIF 2025 base sewing technology** — the specific engineering implementation with optimised looper geometry, synchronisation system and cam thread take-ups.
2. **Automatic thread-trimming device** — the under-plate mechanism with automatic bottom thread clamping and mechanical seam-end locking.
3. **360° circular sewing mechanism** — the special mechanism enabling stitch formation in any direction without rotation of the machine head or lower mechanisms.

Patent Filing Geography

After the 30-month PCT period, national phases will be initiated in priority jurisdictions: USA, China, Germany, Italy, France, Japan, South Korea, India, Bangladesh, Vietnam, Turkey, Russia and EAEU countries, Uzbekistan.

Trade Secrets and Know-How

In addition to patents, critically important knowledge will remain under trade secret: precise component geometry, cam profiles of thread take-ups, calibration methods for specific materials, and synchronisation algorithms. Trade secrets have no expiry date and require no public disclosure.

✓ RESULT:

A competitive barrier for 20 years from the date of patent grant. Even after patents expire, know-how and trade secrets will continue to protect technological leadership.

11.3. Proposal No. 1: Joint Venture for Equipment Manufacturers (50/50)

This proposal is addressed to companies with existing manufacturing infrastructure, engineering teams, distribution networks and market connections — or to companies ready to create such production.

ZARIF 2025 Team Contribution	Manufacturing Partner Contribution
Full technology transfer (patents, know-how, engineering documentation, calibration methods)	All capital investment: production facilities, equipment, tooling, CAD/CAM systems
Technical management, quality certification, ongoing R&D	Full manufacturing responsibility: engineering, production, quality control, supply chain
Brand, intellectual property, international technical support	Distribution network, sales team, customer relations in the country and region
50% of net revenues from machine sales, licensing and service agreements	50% of net revenues from machine sales, licensing and service agreements

Why this is advantageous for the partner:

- Zero technological risk — you receive a fully developed, prototype-validated technology with 5,000+ hours of testing.
- Zero R&D costs — 31 years of investment in development is contributed by ZARIF. You invest in production, not research.

- Instant market leadership — your company becomes the first and only manufacturer of robot-compatible sewing machines in the world.
- Equal upside, limited risk — you build the factory, sell the machines and earn 50% of every sale with the technology contributed free of charge.
- First-mover protection — the joint venture operates under 20-year patent protection.

We are open to collaboration in any country with a developed industrial base: China, Vietnam, India, Turkey, EU countries, USA, Mexico, Brazil. A separate joint venture can be created in each region.

11.4. Proposal No. 2: Round A Funding — $40–50 Million

This proposal is addressed to venture funds, private investors, sovereign funds, AI companies and humanoid robot manufacturers ready to become part of the technological revolution in the sewing industry.

Round Parameter	Value
Round volume	$40–50 million USD
Funding type	Equity participation
Stage	Series A (operating prototype with proven technology)
Objective	From prototype to first industrial machine and serial production
Structure	Single round or syndicate of several investors
Geographic scope	Global (priority — USA, EU, China, Japan, South Korea)

Investment Allocation:

Direction	Volume	What Is Created
Industrial version of the machine (design, prototype, certification)	$8–12M	World's first dual-purpose machine without lubrication
Thread-trimming device (manufacture + 1M cycle testing)	$3–5M	Critical component for 24/7 autonomy
360° circular sewing mechanism (prototype + testing)	$3–5M	Foundation for pattern and specialised automatics
R&D: software, MES, digital twin, AI algorithms	$5–8M	ZARIF 2025 digital platform
Patenting in 20+ jurisdictions via WIPO (PCT)	$2–3M	20-year protective barrier
Production launch and first serial deliveries	$10–15M	Stage I market entry
Operating costs and contingency fund	$9–12M	Operational continuity until profitability

11.5. Comparison with Automation Competitors

Characteristic	SoftWear Automation	Silana / CreateMe	ZARIF 2025
Technological base	Adaptation of traditional machines + machine vision	Narrow specialisation (knitwear) or thread replacement with adhesive	New stitch technology, inherently robot-native
Product versatility	Simple single-layer items only	Limited by material	Full range: from silk to 8 mm leather, material combinations
Continuity of operation	Interrupted by bobbin (every 12–20 min)	Depends on technology	24/7 without stoppages (no bobbin, no breaks)

Gradual implementation	Impossible (complete line replacement)	Impossible	Built in: "human" → "robot" mode on the same machine
IP protection	Patents on automation	Patents on method	Patent on basic stitch principle (US No. 6,095,069) + 3 new PCT applications
Cell cost	$100,000–350,000	N/A	$28,000–36,000 (machine + humanoid robot)
Payback period	5–8 years	N/A	11.1 months

11.6. Why Traditional Manufacturers Cannot Compete

Large sewing equipment manufacturers (Juki, Brother, Jack, Typical, Pfaff, Dürkopp Adler) possess production facilities and distribution networks. However, they face three fundamental barriers that make creating an equivalent to ZARIF 2025 practically impossible:

1. **Technological barrier.** All their technologies are based on the lockstitch (type 301) or traditional chain stitch with an eye-looper (type 401). Transition to the rotary looper requires not modification but a complete architectural change from scratch. 28 years of ZARIF development cannot be replicated quickly.
2. **Patent barrier.** US Patent No. 6,095,069 and three new PCT applications create a legal wall. Even if a competitor wished to develop an equivalent, they would face the impossibility of circumventing the patent protection.
3. **Economic barrier.** Existing manufacturers have multi-billion-dollar investments in lockstitch infrastructure: spare parts stocks, dealer networks, production lines. A revolutionary technology would instantly devalue these assets. This is the classic "innovator's dilemma": market leaders have no economic incentive for a breakthrough that will destroy their current business.

COMPETITIVE VACUUM:

It is precisely the combination of technological, patent and economic barriers that creates a unique window of opportunity for ZARIF 2025. The first to enter the market with a robot-native sewing platform will secure a 20-year competitive advantage.

11.7. A Call to Action: The Time Is Now

Over 31 years, Zarif Tadjibaev answered the question that everyone considered absurd, and spent that time proving that the impossible is possible. Today this technology is ready to transform an industry worth $2.36 trillion.

To equipment manufacturers: You have factories, machine tools, engineers and distributors. You have clients who are awaiting innovation. You lack only one thing — the technology that will change the rules of the game. ZARIF 2025 is that technology. Let us create a joint venture and become leaders of a new era.

To investors: You are seeking opportunities with 10–100x growth potential. You invest in deep technologies that create new markets. The market for autonomous sewing production is only just forming, and ZARIF 2025 is its foundation. Your investments today will become the basis for an industry that in 10 years will be measured in tens of billions of dollars.

To AI companies and robot manufacturers: You have created remarkable machines that walk, see and think. But they cannot sew. Not because they are insufficiently advanced, but because the world has not given them the right tool. ZARIF 2025 is that tool. Let us together create an ecosystem where robots and sewing machines speak the same language (ROS 2), exchange data in real time and work as a single organism.

CONTACT DETAILS FOR NEGOTIATIONS

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CONCLUSION. THREAD AS THE FOUNDATION OF AN AUTONOMOUS FUTURE

We have reached the end of this journey — from the question asked in a Tashkent workshop in 1994, to the technology that in 2025 proved: the ideal single-needle sewn seam is possible. Possible without a bobbin. Possible without an eye-looper. Possible without the compromises that the sewing industry had been making for 170 years.

ZARIF 2025 technology is not an improvement on the old paradigm. It is its complete replacement. Ten breakthroughs, confirmed by 28 years of testing on a worn prototype, define a new system of coordinates in which:

- **Continuity of operation ceases to be a dream** — the bottom thread is supplied from a 1–5 kg cone, and the machine sews 24/7 without interruption.
- **Universality ceases to require retooling** — a material range from 0.1 mm to 8 mm on a single setting.
- **Reliability ceases to depend on micron tolerances** — 0.5 mm clearance, wide tolerances, operation on worn equipment.
- **Quality ceases to be a compromise between elasticity and strength** — the new type of double thread chain stitch combines both, with a reverse side visually identical to the lockstitch.
- **Robotisation ceases to encounter mechanical barriers** — ZARIF 2025 was designed as a robot-native device, speaking the language of ROS 2 and OPC UA.

What ZARIF 2025 Means for Different Industry Participants

For clothing and footwear manufacturers — a sixfold reduction in operating costs, 97% output growth, payback on additional investment in 11.1 months and the possibility of gradually transitioning to fully autonomous production without equipment replacement.

For sewing equipment manufacturers — the chance to become the leader of a new technological era, gaining exclusive access to a platform protected by patents for 20 years and heading a market that will grow to $400–900 billion within 15 years.

For humanoid robot and AI developers — the missing link that transforms a humanoid robot from a laboratory prototype into a fully-fledged production operator. ZARIF 2025 is the killer application for industrial robotics.

For investors — the opportunity to enter a deep technology at the working prototype stage, with protected intellectual property, proven market demand and 10–100x growth potential.

For humanity — the liberation of millions of people from monotonous, physically arduous labour in sewing shops, and the creation of infrastructure for local, sustainable, digital clothing production — closer to the consumer, with a smaller carbon footprint and zero overproduction.

> *170 years of waiting. 31 years of development. The moment has arrived.*
>
> *When Elias Howe invented the lockstitch in 1846, the world was different. Electricity was only beginning its journey and computers were unimaginable. Today, in 2026, artificial intelligence manages factories, humanoid robots assemble vehicles, and satellite internet connects continents. But the sewing machine — the very same that joins the fabric from which our clothing, footwear, car seats and medical masks are made — has until now operated on the principles of 1846.*
>
> *ZARIF 2025 changes this. For ever.*
>
> *Thread — the most ancient joining material in human history — finds new life. Not as a museum exhibit, but as the foundation of autonomous, intelligent and sustainable production for the future. Thread has not become obsolete. It was simply awaiting the right technology. Now it exists.*
>
> *"This technology is not evolution. It is revolution. And it begins today."*
>
> *— Zarif Tadjibaev, Tashkent, 2026*

APPENDICES

Appendix A. Technical Specifications of the ZARIF 2025 Prototype

Parameter	Value
Base stitch type	Double thread chain stitch, new type (loops rotated 180°)
Maximum speed	5,000 stitches/min
Stitch length range	0.5–5 mm
Maximum material thickness	8 mm (needle bar stroke 32 mm)
Needle type	Standard single-groove (DPx5, Groz-Beckert)
Needle size range without readjustment	Nm 60/8 – Nm 130/21
Needle–looper clearance	up to 0.5 mm
Thread supply	Continuous from cones (top and bottom)
Loop tightening modes	Strong / Moderate / Light
End-of-seam tacking	0.5 mm stitches (6–10 stitches) + mechanical lock
Needle plate	With technological slot

Thread take-ups	Rotating cam (two independent, for top and bottom thread)
Condition at time of testing	Worn (play, enlarged clearances after 28 years of experiments)

Appendix B. Target Parameters of the Industrial Machine with Flatbed Platform (Stage I)

Parameter	Target Value
Type	Single-needle, flatbed platform, bottom feed
Maximum speed	5,000+ stitches/min
Stitch length range	0.5–5 mm (optionally up to 10 mm)
Maximum material thickness	8 mm (stroke 32 mm)
Thread supply	Continuous from cones (up to 1–5 kg)
Tension control	Digital, stepper motors
Automatic functions	Thread trimming, tacking, presser foot lift, needle positioning
Auto-trimming system	With mechanical lock, service life 1 million cycles
Operator interface	7–10 inch touch display, voice control
Robot interfaces	ROS 2, OPC UA, MQTT, API
Operating modes	Human operator (pedal) / Humanoid robot
Expected price	$5,000–6,000

Appendix C. Industrial Version Bill of Materials (BOM) ◈

Assembly	Component	Material	Note
Main shaft	Bearings	Si_3N_4 + 100Cr6 steel with DLC coating	Class P5, dry operation
Main shaft	Shaft	ШХ15 steel, hardened 58–62 HRC	—
Needle bar	Bushings	PEEK-CF30 (30% carbon fibre)	Ra 0.2 μm, dry friction
Connecting rod	Body	Aluminium 7075-T6, anodised	–70% by mass
Connecting rod	Lower head	Iglidur X (self-lubricating)	No liquid lubrication
Guide block	Guides	POM-C with graphite	Quiet operation without oil
Thread take-up	R45 discs	Sheet carbon (CF) 1 mm	Laser cut, polished
Thread take-up	Fork rods	Stainless SUS316, Ø2 mm	Electropolished
Looper	Monolithic	Tool steel + TiN coating	Hardened 60–62 HRC
Covers	Acoustic	ABS + foamed polyurethane inside	Sound absorption
Covers	Transparent door	Impact-resistant polycarbonate (Lexan)	Anti-static coating

Appendix D. Electronic Components of the Industrial Version ◈

Component	Type / Model	Purpose
Main servo motor	BLDC Direct Drive 750 W Active Torque	Needle positioning

Presser foot pressure motor	NEMA 17 stepper + lead screw	Digital presser foot pressure
Thread tension motors	2 × NEMA 8 (20×20×40 mm)	Digital top and bottom thread tension
Material height sensor	Magnetic AS5311 (10×20 mm)	Precision 0.01 mm
Top thread break sensor	Piezoceramic (15×20×40 mm)	Inside transparent cover
Bottom thread break sensor	Piezoceramic (10×10×5 mm)	In platform near looper
Mini-supercomputer	NVIDIA Jetson Orin NX (8 GB, 20 TOPS)	AI, machine vision
Cameras	2 × 4K high-speed	Above needle bar and in front of presser foot
Controller	STM32H7 or equivalent	CAN FD, EtherCAT
HMI display	Touch LCD 7–10 inches	Preset management

Appendix E. US Patent No. 6,095,069 — Key Provisions ✓

The patent was granted on 1 August 2000 in the name of Zarif Sharifovich Tadjibaev. Full text available at: https://patents.google.com/patent/US6095069A

Abstract: A device for forming a double thread chain stitch with a rotary looper. The looper makes two rotations per needle cycle. The invention eliminates the eye-looper, simplifies the mechanism and increases reliability.

Appendix F. Glossary of Principal Terms

Term	Definition
Type 301	Lockstitch. Thread interlacing inside the material. Requires a bobbin.
Type 401	Double thread chain stitch. Thread interlacing on the underside of the material.
Rotary looper	A looper performing continuous rotation. In ZARIF 2025 — 2 rotations per cycle. Has no eye.
Top thread	The thread passing through the needle eye; forms the stitch from above.
Bottom thread	The thread supplied from the cone below; forms the stitch from beneath.
Lockstitch	See Type 301. The interlaced stitch where threads lock inside the material.
Double thread chain stitch	See Type 401. Stitch formed by two threads interlacing below the material.
Eye-looper	A looper with an eye through which the bottom thread is threaded; used in traditional type 401.
Tacking (condensation stitch)	A group of very short stitches (0.5 mm) at the beginning and end of a seam to prevent unravelling.
Thread take-up	The mechanism that feeds and tightens the thread during stitch formation.
DLC coating	Diamond-Like Carbon — diamond-like carbon. Hardness >2,500 HV, dry friction.
PEEK	Polyether ether ketone — a high-temperature engineering polymer replacing metal.
Iglidur	Self-lubricating Igus polymers for operation without liquid lubrication.
ROS 2	Robot Operating System 2 — standard middleware for humanoid robots.
OPC UA	Industrial standard for data exchange between equipment and MES/ERP.

MES	Manufacturing Execution System — production management system.
SLAM	Simultaneous Localisation and Mapping — AGV trolley navigation.
Lights-out manufacturing	Production without personnel on the floor — "lights off."
Active Torque	Maintenance of servo motor torque under varying load.
OEE	Overall Equipment Effectiveness.
AGV	Automated Guided Vehicle — automatic transport vehicle.
Dry Head	Sewing head architecture without liquid lubrication.
Zero-Config	The machine's ability to sew different materials without mechanical adjustment.

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If you are interested in investing, joint development, licensing or strategic partnership, please contact the author to discuss opportunities.

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END OF BOOK

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